Periodic Table

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Periodic Table

Introduction:
It is quite natural to begin your study of inorganic chemistry with the study of the periodic table in this unit.
Periodic table helps us to undergo a systematic study of the various elements found in nature, without which it would have been impossible for us to study all the elements.

Q1 **Why do we need to classify elements?**

Sol:

Year	Number of elements discovered
1800	31
1865	63
At present	118

It is very difficult to study all the properties of such large number of elements individually. So scientists searched for a systematic way to organise these elements.

Q2 **What is periodic table?**

Sol: Periodic Table is a systematic arrangement of all the elements known to us in accordance with their increasing atomic number and recurring chemical and physical properties. They are arranged in tabular form to keep similar property elements in vertical column (known as groups). Horizontal rows are known as periods.

Genesis of Periodic Classification: -
(i) Dobereiner's Law of Triads:
He formed group of 3 elements in such a way that, when the elements are arranged according to increasing atomic mass, then atomic mass of middle element is average of atomic masses of first and last element.
He found that elements of such groups have similar properties. Such groups were known as Dobereiner's Triad.
Mathematically: If elements A, B, C have atomic masses x, y, z respectively, then $\left(y = \frac{x+z}{2}\right)$, also A,B,C have similar properties.

e.g.: Li(A=7), Na(A=23), K(A=39)

atomic mass of Na = 23 = $\frac{7+39}{2}$ = average of atomic masses of Li and K

More examples of Dobereiner's Triad:

Ca(A=40), Sr(A=88), Ba(A=137)
Cl(A=35.5), Br(A=80), I(A=127)
P(A=31), As(A=75), Sb(A=120)
S(A=32), Se(A=79), Te(A=127)

Trick : If atomic numbers of elements A,B and C are x,y and z , and (y-x)= (z – y), then A,B and C are in triad. (no need of atomic mass).

e.g.: $\underbrace{Li(z=3), Na(z=11), K(z=19),}_{\text{Dobereiner's Triad}} \Rightarrow 11-3=19-11=8$

e.g.: $\underbrace{Ca(z=20), Sr(z=38), Ba(z=56)}_{\text{Dobereiner's Triad}} \Rightarrow 38-20=56-38=18$

Drawbacks:

He wasn't able to classify all the known elements into triads.
Properties of elements of different triads can't be compared with each other.
He made some triads based on similarities of properties but do not follow average of atomic mass rule, such as:-

(i) H(A=1), Li(A=7), Na(A=23)

Average = $\frac{23+1}{2}$ = 12 ≠ atomic mass of Li

(ii) O(A = 16), S(A = 32), Se(A = 79)

Average= $\frac{16+79}{2}$ = 47.5 ≠ atomic mass of S

Q3 **Atomic weight of an element x is 39, and that of element Z is 132, an atomic weight of their intermediate element Y, as per Dobereiner's Triad, will be?**
(A) 88.5 **(B) 93.0** **(C) 171** **(D) 85.5**

Sol: **D**

Given elements $\underset{39}{X}\ \underset{?}{Y}\ \underset{132}{Z}$

According to Dobereiner's Triad = $\frac{X+Z}{2}$ = Y

$$= \frac{39 + 132}{2}$$

$$= \frac{171}{2}$$

$$= 85.5$$

So, Atomic weight of Y = 85.5

Q4 **The law of triads is applicable to a group:**
(A) C, N, O **(B) H, O, N** **(C) Na, K, Rb** **(D) Cl, Br, I**

Sol: **D**

For Cl, Br, I

$$\frac{35.5 + 1269.9}{2} = 81.2 \simeq 80$$

So, the law of trades is applicable only cl, Br, I

Q5 **Predict the approximate density of Cs from the following data using law of triad. K = 0.868 g cm^{-3}; Rb = 1.532 g cm^{-3}; Cs = ?**

Sol: $\frac{d_{Cs} + d_K}{2} = d_{Rb}$

$\because$ Atomic number of K = 19, Rb = 37 Cs = 55

$\therefore \quad d_{Cs} = 2d_{Rb} - d_K$

$= 2 \times 1.532 - 0.868 = 2.196$

Q6 **The b.pt. of Kr and Rn are –152°C and -62°C respectively. Calculate the approximate b.pt, of Xe.**

Sol: $\text{b.pt. of Xe} = \frac{\text{b.pt. of Kr} + \text{b.pt. of Rn}}{2}$

$= \frac{-152 - 62}{2} = -107^0\text{C}$

(ii) A.E.B. de Chancourtois:
He attempted to classify elements in order of increasing atomic weight and made a cylindrical table which was used to display periodic recurrence of properties.
Drawback: He only published his paper but did not-publish his actual graph with irregular arrangements

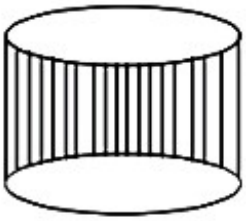
Q7 **The telluric helix was given by:**

(A) Dechan Courtois **(B) Newlands**

(c) L. Meyer **(D) Mendeleev**

Sol: **A**

Telluric helix was gives by Dechan Courtois .
Telluric helix is 16 equal lines and elements arrange in atomic weight



(iii) Newland's Law of Octaves:
He arranged the elements in increasing order of their atomic weights and observed that every eighth element had properties similar to the first element just like every eighth note that resembles the first note in octaves of music.

Do	**Re**	**Mi**	**Fa**	**So**	**La**	**Ti**
Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl
K	Ca	- - -	- - - - - - - -		- - - - -	

Drawbacks:
His Law seemed to be true only for elements upto calcium.
Until then noble gases weren't discovered. After discovery of noble gases, his law was modified. That is, properties of every ninth element are similar to properties of first one.

(iv) Lothar Meyer's Curve:

He plotted the physical properties such as atomic volume, melting point and boiling point against atomic weight and obtained a periodically repeated pattern.

He observed that elements having similar properties occupy similar positions.

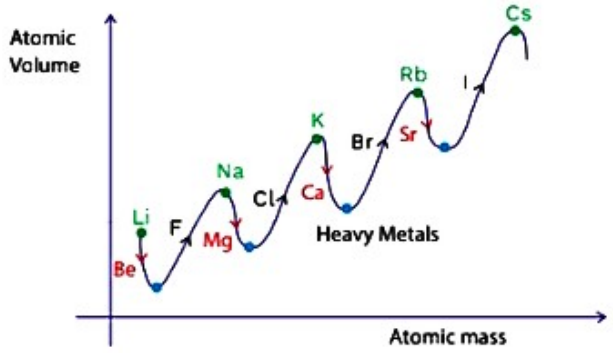
In the following graph of atomic volume vs atomic mass, he observed that :

i) Alkali metals occupy maxima of curve.

ii) Alkaline earth metals occupy midpoints of descending curve.

iii) Halogens occupy midpoints of ascending curve.

iv) Heavy Metals occupy minima of curve.



Drawbacks:

Graphical method was difficult to study.

He also developed a table but before he got it published, Mendeleev's Periodic Table was already published.

It was difficult to find similar positions on a curve of different properties as periodicity and nature of graph was not similar.

(v) Mendeleev's Periodic Table:

He was responsible for publishing the Periodic Law for the first time. It states as follows:

"Atomic mass is the fundamental property of elements and all other physical and chemical properties are periodic function of atomic mass."

He fully recognised the significance of periodicity and used broader range of physical and chemical properties to classify elements.

His main basis was the similarities in empirical formulas and properties of compounds formed by the elements.

Characteristics:

He arranged 63 known elements into:
12 horizontal rows ⇒ series
8 vertical columns ⇒ Groups.
Groups were further divided:
Subgroup A: Normal Elements.
Subgroup B: Transition elements.
Some of 12 series were grouped horizontally, these horizontal groups were known as periods. (7 periods)
The VIII group consist of 9 elements in three vertical columns.

Merits:

He classified all known elements into groups according to their similar properties. Hence, study of elements was made easier due to tabular classification.
He realised that some elements did not fit in right place according to other properties, if placed according to atomic weight. So he ignored atomic weights, thinking that measurements might be incorrect.
Hence, led to corrections of atomic weights of some elements : : U, Be, In, Au, Pt.
As he arranged elements with similar property, he considered that some elements were still undiscovered, and left several gaps, which helped in discovery of new elements.

Eka	Aluminium(Al)	→	Ga	(Aaloo – Galoo)
Eka	Silicon(Si)	→	Ge	(Sita – Geeta)
Eka	Boron (B)	→	Sc	(B.Sc.)
Eka	Manganese(Mn)	→	Tc	(M.Tech)

Demerits:

Position of Hydrogen was not fixed.
Position of isotopes was not clear.
He could not explain cause of periodicity.
Some pairs of elements do not follow the increasing order of atomic weights, known as Anomalous Pairs.

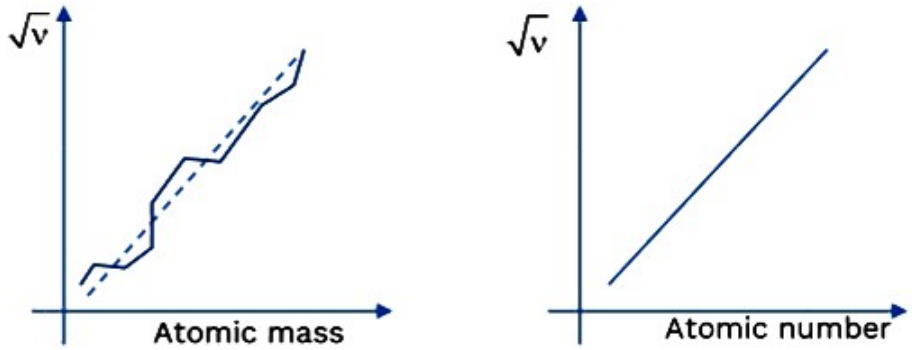
$$\begin{pmatrix} Te, & I \\ 127.5 & 127 \end{pmatrix} \begin{pmatrix} Co, & Ni \\ 58.9 & 58.6 \end{pmatrix} \begin{pmatrix} Th, & Pa \\ 232 & 231 \end{pmatrix}$$

SERIES	GROUPS OF ELEMENTS								
	0	I	II	III	IV	V	VI	VII	VIII
1	-	Hydrogen H 1.008	-	-					
2	Helium He 4.0	Lithium Li 7.03	Beryllium Be 9.1	Boron B 11.0	Carbon C 11.0	Nitrogen N 14.04	Oxygen O 16.00	Fluorine F 19.0	
3	Neon Ne 19.9	Sodium Na 23.5	Magnesium Mg 24.3	Aluminium Al 27.0	Silicon Si 28.4	Phosphorus P 31.0	Sulphur S S32.06	Chlorine Cl 35.45	
4	Argon Ar 38	Potassium K 39.1	Calcium Ca 40.1	Scandium Sc 44.1	Titanium I 48.1	Vanadium V 51.4	Chromium Cr 52.1	Manganese Mn 55.0	Iron Fe 55.9 / Cobalt Co 59 / Nickel Ni 59 / (Cu)
5		Copper Cu 63.6	Zinc Zn 65.4	Gallium Ga 70.0	Germanium Ge 72.3	Arsenic As 75	Selenium Se 79	Bromine Br 79.95	
6	Krypton Kr 81.8	Rubidium Rb 85.4	Strontium Sr 87.6	Yttrium Y 89.0	Zirconium Zr 90.6	Niobium Nb 94.0	Molybdenum Mo 96.0	-	Ruthenium Ru 101.7 / Rhodium Rh 103.0 / Palladium Pd 106.5 / (Ag)
7		Silver Ag 107.9	Cadmium Cd 112.4	Indium In 114.0	Tin Sn 119.0	Antimony Sb 120.0	Tellurium Te 127.6	Iodine I 126.9	
8	Xenon Xe 128	Caesium Cs 132.9	Barium Ba 137.4	Lanthanum La 139	Cerium Ce 140	-	-	-	
9	-	-	-	-	-	-	-	-	
10	-	-	-	Ytterbium Yb 173	-	Tantalum Ta 183	Tungsten W 184	-	Osmium Os 191 / Iridium Ir 193 / Platinum Pt 194.9 / (Au)
11.		Gold Au 197.2	Mercury Hg 200.0	Thallium Tl 204.1	Lead Pb 206.9	Bismuth Bi 208	-	-	
12.	-	-	Radium Ra 224	-	Thorium Th 232	-	Uranium U 239		
	R	R_2O	RO	R_2O_3	HIGHER SALINE OXIDES RO_2 R_2O_5 RO_3 R_2O_7 HIGHER GASEOUS HYDROGEN COMPOUNDS RH_4 RH_3 RH_2 RH				RO_4

Q8 **Which main group elements have a different number of outer most electron than its group number of Mendeleev's periodic table?**

Sol: The noble gases group zero, they have 8 valence electrons.

(Vi) Modern Periodic Law & The Modern Periodic Table:
Henry Mosley's Experiment!
Mosley measured frequency of x-rays emitted by metal, when it is bombarded with high speed e^- and plotted these two graphs:



He observed that the plot of $\sqrt{\nu}$ vs atomic mass is approximately a straight line but $\sqrt{\nu}$ vs atomic number is perfectly a straight line.
So, He concluded that instead of atomic mass, atomic number should be fundamental property.
Neils Bohr considered Mosley's suggestion and modified Mendeleev's periodic law, to give:

Modern Periodic Law:
"Atomic Number is the fundamental property of elements and all other physical and chemical properties are periodic functions of atomic number."

Modern Periodic Table / Long form of Periodic Table :
Modern periodic table is constructed on the basis of modern periodic law and electronic configuration.
It consists of : 18 vertical columns ⇒ Groups
7 horizontal rows ⇒ Periods

Number of elements ⇒ 118.
Orbitals filled in 'n^{th}' period: (ns) ...(n−3)g, (n−2)f, (n−1)d, np

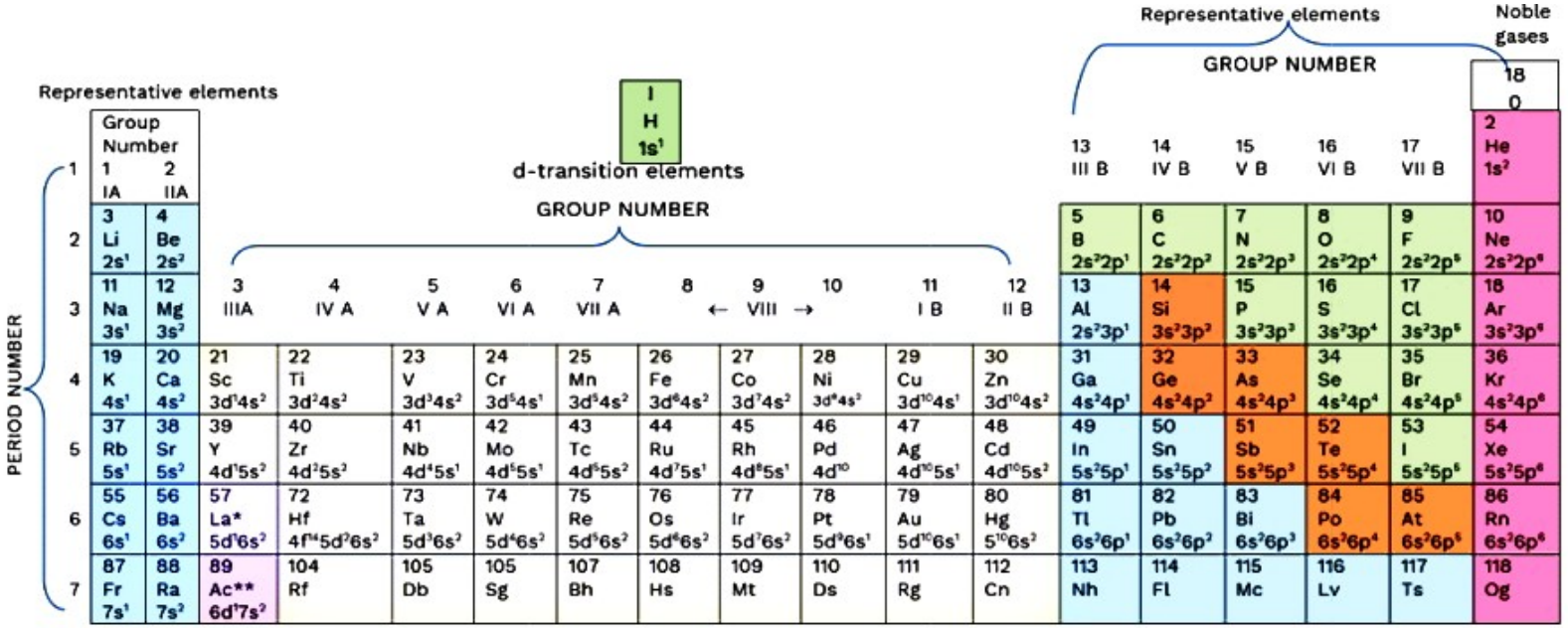
Number of elements in 'n^{th}' period:

Period	orbitals filled	No. of elements	type of period
1	1s	1 × 2 = 2	shortest
2	2s, 2p	4 × 2 = 8	short
3	3s, 3p	4 × 2 = 8	short
4	4s, 3d, 4p	9 × 2 = 18	long
5	5s 4d, 5p	9 × 2 = 18	long
6	6s, 4f, 5d, 6p	16 × 2 = 32	longest
7	7s, 5f, 6d, 7p	16 × 2 = 32	longest
		Magic Nos.↑	

Mathematically:
Number of elements in n^{th} period:

$$= 2\left(\frac{n+2}{2}\right)^2 \text{ If n is even}$$

$$= 2\left(\frac{n+1}{2}\right)^2 \text{ If n is odd}$$



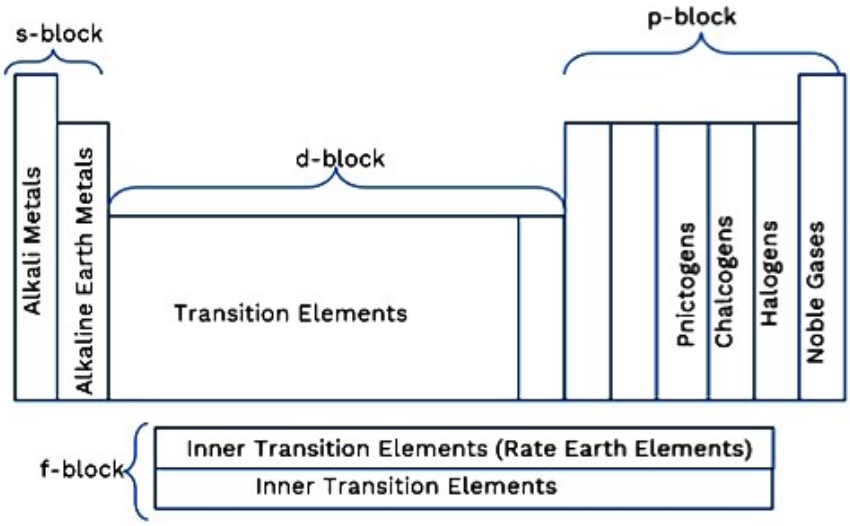
*Lanthanoids $4f^n5d^{0-1}6s^2$	58 Ce $4f^25d^06s^2$	59 Pr $4f^35d^06s^2$	60 Nd $4f^45d^06s^2$	61 Pm $4f^55d^06s^2$	62 Sm $4f^65d^06s^2$	63 Eu $4f^75d^06s^2$	64 Gd $4f^75d^16s^2$	65 Tb $4f^95d^06s^2$
	66 Dy $4f^{10}5d^06s^2$	67 Ho $4f^{11}5d^06s^2$	68 Er $4f^{12}5d^06s^2$	69 Tm $4f^{13}5d^06s^2$	70 Yb $4f^{14}5d^06s^2$	71 Lu $4f^{14}5d^16s^2$		
**Actinoids $5f^n6d^{0-2}7s^2$	90 Th $5f^06d^27s^2$	91 Pa $5f^26d^17s^2$	92 U $5f^36d^17s^2$	93 Np $5f^46d^17s^2$	94 Pu $5f^66d^07s^2$	95 Am $5f^76d^07s^2$	96 Cm $5f^76d^17s^2$	97 Bk $5f^96d^07s^2$
	98 Cf $5f^{10}6d^07s^2$	99 Es $5f^{11}6d^07s^2$	100 Fm $5f^{12}6d^07s^2$	101 Md $5f^{13}6d^07s^2$	102 No $5f^{14}6d^07s^2$	103 Lr $5f^{14}6d^17s^2$		

f-Inner transition elements

Trick

Group 1:	He	Li Na	k	Rb	Cs	Fr					
	He	Lina	ki	Rab	Se	fariyaad					
Group 2:	Be	Mg	Ca	Sr	Ba	Ra					
	Beta	Mange	Car	Scooter	Bapu	Raji					
Group 3:	Sc	y	La	Ae							
	Scooter	Yamaha	Lao	Esi							
Group 4:	Ti	Zr	Hf								
	Tea	Jara	half								
Group 5:	V	Nb	Ta	Group 6:	Cr	Mo	w				
	Vaah	Nobita			Corona	Maut	Wala				
Group 7:	Mn	Tc	Re	Group 8:	Fe	Ru	Os				
	Man	Tak	Rahe		Ferrous						
Group 9:	Co	Rh	Ir	Group 10:	Ni	Pd	Pt				
	Courier				Nahi	padoge pitoge					
Group 11:	Cu	Ag	Au	Group 12:	Zn	Cd	Hg				
	kyu	Aage	Aau		zindagi	Cadbury	Hogi				
Group 13:	B	Al	Ga	In	Tl						
	Bengain	Aaloo	Gajar	in	Thela						
Group 14:	C	Si	Ge	Sn	Pb						
	Chemistry	sir	gives	sanki	problem						
Group 15:	N	P	As	Sb	Bi						
	Nana	Patekar	Asraani	sab	Bimar						
Group 16:	O	S	Se	Te	Po						
	Oye	Sanam	Se	Tera	Pol	Khol dunga					
Group 17:	F	Cl	Br	I	At						
	Full	Colour	Brightness in		Akai TV						
Group 18:	He	Ne	Ar	Kr	Xe	Rn					
	Hema	Neelam	aur	Kareena	ka x-ray	Rangeen					
3d-Series:	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	
	Science	Teacher	Vinita	Kumari	Mann	Fe	Ko	Ni	Kyu	Jaan Jalati ho	
4d-series:	y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	
	Yaari	zara	Nibha	Maut	Tak	Rukavate Rah main Pade to			Aag me	Cud Ja	
5d-series:	La	Hf	Ta	W	Os	Ir	Pt	Au	Hg		
	Laa	Hafta		verna	Osama se Idhar		Pitai	Aur	Hogi		
6d-Series:	AC	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	
	AC main Betha Ruther ford dub gaya sagar main bhai Has Mat, dosa Kha raha Rahul Gandhi Kone Main										
4f-Series:	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
	Sir	par Nadiyan Prem Ki samayi ya gadgad tab dil hua are Tumhe to Yubhi loot lu												
5f-Series:	Th	Pa	U	Np	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	
	Tumhe padau newspaper purana, Aam Kam Bike, Café main aish farmate Modona-Lara													

Modern Periodic Table can be classified into four blocks:



(1) S-block: Elements in which last e^- enters in s-subshell of valence shell, are placed in S-block

1 IA	2 IIA
3 Li $2s^1$	4 Be $2s^2$
11 Na $3s^1$	12 Mg $3s^2$
19 K $4s^1$	20 Ca $4s^2$
37 Rb $5s^1$	38 Sr $5s^2$
55 Cs $6s^1$	56 Ba $6s^2$
87 Fr $7s^1$	88 Ra $7s^2$

Alkali Metals

Alkaline Earth Metals

→ General electronic configuration : ns^{1-2}

→ General oxidation state of group-1 and group-2 is +1 and +2 respectively.

→ Larger in size.

→ Soft metals and silvery white in colour

→ Mostly forms ionic compound.

→ Have low IP and highly reactive.

so, never found in pure form.

→ strong reducing agents

→ Gives flame test (except Be and Mg).

(2) p-block: Elements in which last e⁻ enters in p-subshell of valence shell, are placed in p-block.

General E.C. $\rightarrow ns^2\ np^{1-6}$

$ns^2 np^1$	$ns^2 np^2$	$ns^2 np^3$	$ns^2 np^4$	$ns^2 np^5$	$ns^2 np^6$
13	14	15	16	17	18
B	C	N	O	F	Ne
Al	Si	P	S	Cl	Ar
Ga	Ge	As	Se	Br	Kr
In	Sn	Sb	Te	I	Xe
Tl	Pb	Bi	Po	At	Rn
Nh	Fl	Mc	Lv	Ts	Og

→ s-block elements and p-block elements together known as Representative elements or Main Group Elements.

→ Good oxidising agents

→ Mostly forms covalent bonds.

→ Harder and heavier than s-block elements and have high EA.

p-block elements are mostly non-metals and metalloids but few are Metals.
(Metalloids →Ge, As, Se, Te, Sb)
H, N, 0, F, Cl and noble gases exist in gaseous form at room temperature.
Br is the only non-metal which exist in liquid form.
Grp 15 - Pnictogens: refers to choking / suffocating property
Grp 16 - Chalcogens: ore forming elements.
Grp 17 - Halogens: sea-salt forming elements.
Grp 18 - Noble / Inert gas: Chemically inert elements.

(3) d-block:- Elements in which last e⁻ enters in d-subshell of penultimate shell are placed in d-block.

	3 IIIA	4 IV A	5 V A	6 VI A	7 VII A	8 ←	9 VIII	10 →	11 I B	12 II B
1st Transition series 3d series	21 Sc $3d^1 4s^2$	22 Ti $3d^2 4s^2$	23 V $3d^3 4s^2$	24 ☆ Cr $3d^5 4s^1$	25 Mn $3d^5 4s^2$	26 Fe $3d^6 4s^2$	27 Co $3d^7 4s^2$	28 Ni $3d^8 4s^2$	29 ☆ Cu $3d^{10} 4s^1$	30 Zn $3d^{10} 4s^2$
2nd Transition series 4d-series	39 Y $4d^1 5s^2$	40 Zr $4d^2 5s^2$	41 ☆ Nb $4d^4 5s^1$	42 ☆ Mo $4d^5 5s^1$	43 Tc $4d^5 5s^2$	44 ☆ Ru $4d^7 5s^1$	45 ☆ Rh $4d^8 5s^1$	46 ☆ Pd ☆ $4d^{10}$	47 ☆ Ag $4d^{10} 5s^1$	48 Cd $4d^{10} 5s^2$
3rd Transition series 5d-series	57 ☆ LA* $5d^1 6s^2$	72 Hf $4f^{14} 5d^2 6s^2$	73 Ta $5d^3 6s^2$	74 W $5d^4 6s^2$	75 Re $5d^5 6s^2$	76 Os $5d^6 6s^2$	77 Ir $5d^7 6s^2$	78 ☆ Pt $5d^9 6s^1$	79 ☆ Au $5d^{10} 6s^1$	80 Hg $5d^{10} 6s^2$
4th Transition series 6d-series	89 ☆ Ac** $6d^1 7s^2$	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 ☆ Rg	112 Cn

☆ Exceptions of (n+l) rule

not Transition Elements

General electronic configuration $ns^2(n-1)d^{0-10}$

All d-block elements except group-12 elements are known as transition elements.

Transition Elements: Elements with incomplete d-subshell of penultimate shell, either in ground state or stable oxidation state.

All d-block elements are hard metals, hence show all properties of metals - good conductor of heat and electricity, malleable, ductile, sonorous etc.

They have variable oxidation state due to less energy gap between (n-1)d and ns subshells.

They mostly form coloured ions and show paramagnetism.

Coinage Metals – Cu, Ag, Au

Noble Metals – Au, Pt

(4) f-block: Elements in which last e⁻enters in f-subshell of anti-penultimate shell, are placed in f-block.

* Lanthanoids $4f^{1-14}5d^{0-1}6s^2$	58 Ce ☆ $4f^1 5d^1 6s^2$ ☆	59 Pr $4f^3 5d^0 6s^2$ ☆	60 Nd $4f^4 5d^0 6s^2$ ☆	61 Pm $4f^5 5d^0 6s^2$ ☆	62 Sm $4f^6 5d^0 6s^2$	63 Eu $4f^7 5d^0 6s^2$	64 Gd ☆ $4f^7 5d^1 6s^2$ ☆	65 Tb $4f^9 5d^0 6s^2$	66 Dy $4f^{10} 5d^0 6s^2$	67 Ho $4f^{11} 5d^0 6s^2$	68 Er $4f^{12} 5d^0 6s^2$	69 Tm $4f^{13} 5d^0 6s^2$	70 Yb $4f^{14} 5d^0 6s^2$	71 Lu $4f^{14} 5d^1 6s^2$
** Actinoids $5f^{1-14}6d^{0-2}7s^2$	90 ☆ Th $5f^0 6d^2 7s^2$	91 Pa $5f^2 6d^1 7s^2$	92 U $5f^3 6d^1 7s^2$	93 Np $5f^4 6d^1 7s^2$	94 Pu $5f^6 6d^0 7s^2$	95 Am $5f^7 6d^0 7s^2$	96 Cm $5f^7 6d^1 7s^2$	97 Bk $5f^9 6d^0 7s^2$	98 Cf $5f^{10} 6d^0 7s^2$	99 Es $5f^{11} 6d^0 7s^2$	100 Fm $5f^{12} 6d^0 7s^2$	101 Md $5f^{13} 6d^0 7s^2$	102 No $5f^{14} 6d^0 7s^2$	103 Lr $5f^{14} 6d^1 7s^2$

☆Exceptions of (n+l) rule

All f-block elements belongs to group-3, hence group - 3 is longest group with 32 elements.

General electronic configuration $\rightarrow ns^2(n-1)d^{0-1}(n-2)f^{1-14}$

These are also known as Inner Transition Elements, as they are positioned between transition elements.

'Pm' among Lanthanoids and all Actinoids are radioactive.

Elements having atomic no. (Z) greater than U (Z = 92), are man-made and termed as Transuranic elements.

General stable oxidation state of f-block elements is +3, but lanthanides also show +2 & +4, while Actinides show +3, +4, +5, +6, +7.

Lanthanoids are also known as rare earth elements.

Metals, Non-Metals and Metalloids:

Metals	Non-Metals
Metals are hard and are usually solid at room temperature (Hg is liquid at 25°C)	They are usually soft and gas or solid at room temperature (Br is liquid at 25°C)
They are malleable and ductile	They are brittle

Metals	Non-Metals
They are sonorous	They are non-sonorous
Good conductors of heat and electricity	They are bad conductors (except : graphite)
They have high melting and boiling point	They have low melting and boiling points
High tensile strength	Low tensile strength

Metallic character in periodic table
In period : Left to right decreases
In group: top to bottom increases
Metalloids : Elements that show properties of both metals as well as non-metals are metalloids or Semi-Metals.
Positioned between metals and non-metals. (Ge, As, Se, Te, Sb)

Q9 **How many elements are yet to be discovered to complete the last period of the periodic table? What shall be the atomic number of the last member of the last period and to which group it shall belong?**

Sol: 13 elements, 118, zero group

Q10 **Why are the elements having ns^2, np^6 configuration called inert?**

Sol: The elements having ns^2, np^6 configuration have complete octet and thus neither possess the tendency to donate electron nor to accept electron and are thus stable.

Q11 **There are only 14 lanthanoids and only 14 actinoids. Why?**

Sol: Lanthanoids and actinoids are the elements in which differentiating electron enters in to (n-2) f-subshell. Since 'f' subshell can accommodate a maximum of 14 electrons and thus number of lanthanoids and actinoids are 14.

Q12 **3d, 4d and 5d series consists of 10 elements each. Explain.**

Sol: In 3d, 4d and 5d series, the differentiating electron enters to (n-1) d-subshell. The maximum capacity of d-subshell is of 10 electrons. Thus, there are 10 elements each in 3d, 4d and 5d series [$(n-1)\ d^{1\text{-}10}$].

Q13 **There are 2, 8 and 8 elements in first, second and third periods of periodic table respectively. Explain.**

Sol: In first period 1s is completed. Its capacity is of two electrons. In second period 2s 2p and in third period 3s 3p are completed. The capacity of these shells is of 8 electrons each. Thus, 2, 8 and 8 elements are present in first, second and third periods respectively.

Q14 **What is the atomic number of the element having maximum number of unpaired 2p electrons. To which group it belong?**

Sol: ,N, it belongs to 15 (or V A) group.

Q15 **In which subshell does the differentiating electron enters in case of inner transition element?**

Sol: (n-2) f-subshell.

Q16 **Why is argon (at. wt. 39.94) placed before potassium (at. wt. 39.10) in the periodic table?**

Sol: The periodic table is based on atomic number and the atomic number of Ar is 18 which comes first as the at. no. of K is 19 .

Q17 **Boron is metalloid. Comment.**

Sol: It shows the metallic as well as non-metallic nature.

Q18 **Sodium is strongly metallic, while chlorine is strongly non-metallic. Explain.**

Sol: Sodium (ns^1) has low IE and thus possesses strong tendency to lose electron and exhibits strong metallic nature, On the other hand chlorine (ns^2 np^5) possesses high electron affinity and thus shows more tendency to accept electron to show strong non-metallic nature.

Q19 **Why are metals good conductor of current?**

Sol: Metals have low ionisation energies and so valence electrons form an ocean of loosely bonded electrons. These loosely bonded electrons are responsible for conduction of current in metals.

Q20 **Why are inert gases mono-atomic?**

Sol: Inert gases have complete octet (ns^2, np^6) and so do not form bonds.

Q21 **Why are the f-block elements so called?**

Sol: The differentiating electron enters to in (n-2) f-subshell and thus elements are called 'f-block elements.

Q22 **Why are s-block elements most reactive?**

Sol: Because of low IE values, s-block elements possess more tendency to lose electron and are thus more reactive.

Q23 **Why are zero group elements called noble gases or inert gases?**

Sol: These possess ns^2 np^6 configuration and are thus chemically inert.

Q24 **How many elements out of first 100 elements possess 3d electrons?**

Sol: 80 (since filling of 3d-orbital starts from elements with at. no. 21).

Q25 **By examining the modern periodic table, find as many examples as you can of the violation of Mendeleev's periodic law that the chemical and physical properties of elements are periodic function to their atomic weights.**

Sol: Exceptions to Mendeleev's periodic law: CO and K ; Co and Ni, Te and I: Th and Pa: U and Np: Pu and Am. If placed in order of increasing atomic weight, these elements would not fall in the group with which they have similar properties.

Q26 **Why is the third period contains 8 elements and not 18 ?**

Sol: The filling of electrons in third period begins from $3s^1$ and completes at $3p^6$.

Q27 **Arrange the following pure solid elements in order of their increasing electrical conductance at room temperature: Ge, Ca, S and Si : Also explain the reason.**

Sol: S < Si < Ge < Ca ; Metals are good conductor of current and non-metals are poor ones. Metallic character decreases from left to right across a period and from bottom to top in a group.

Q28 **Arrange the following elements in order of increasing melting point and give reason.**
Al, F, Na and Mg.

Sol: F < Na < Mg < Al ; Fluorine is non-metal and exists as a diatomic gas, so it has lowest melting point. Na, Mg and Al are metals in the third period; their melting point should increase as the metallic character decreases from left to right in the period.

Q29 **Why do different periods of the periodic table have different number of elements?**

Sol: Each shell has different number of subshells and each period starts with filling of new shell and completes when it occupies ns^2 np^6 configuration.

Q30 **Point out the most important relationship among elements of the same group in periodic table.**

Sol: Each has same valence shell configuration.

Q31 **What would be the atomic number of alkaline earth element if it is discovered and belongs to 8th period.**

Sol: 120

Q32 **Lanthanoids and actinoids are placed in separate rows at the bottom of the Periodic Table. Explain the reason for this arrangement.**

Sol: These element having incomplete f-orbitals and show properties which are distinct from all other elements.

IUPAC Naming of Elements with atomic number > 100:

Digit	0	1	2	3	4	5	6	7	8	9
Nomenclature	nil	un	bi	tri	quad	pent	hex	sept	oct	enn
Symbol	n	u	b	t	q	p	h	s	o	e

Atomic No.	IUPAC Name	Symbol	Elemental Name	Symbol
101	Unnilunium	Unu	Mendelevium	Md
102	Unnilbium	Unb	Nobelium	No
103	Unniltrium	Unt	Lawrencium	Lr
104	Unnilquadium	Unq	Rutherfordium	Rf
105	Unnilpentium	Unp	Dubnium	Db
106	Unnilhexium	Unh	Seaborgium	Sg
107	Unnilseptium	Uns	Bohrium	Bh
108	Unniloctium	Uno	Hassium	Hs
109	Unnilennium	Une	Meitnerium	Mt
110	Ununnillium	Uun	Darmstadtium	Ds

Atomic No.	IUPAC Name	Symbol	Elemental Name	Symbol
111	Unununnium	Uuu	Rontgenium	Rg
112	Ununbium	Uub	Copernicium	Cp
113	Ununtrium	Uut	Nihonium	Nh
114	Ununquadium	Uuq	Flerovium	Fl
115	Ununpentium	Uup	Moscovium	Mc
116	Ununhexium	Uuh	Livermorium	Lv
117	Ununseptium	Uus	Tennessine	Ts
118	Ununoctium	Uuo	Oganesson	Og

Group, period, Block identification:

(a) If electronic configuration is given:
Block: given by subshell in which last e^- enters.
Period: largest principle quantum no.
Group: If it belongs to 'n^{th}' period and
s-block, then group no. = no. of ns e^-
p-block, then group no. = 10 + no. of (ns + np)e^-
d-block, then group no. = no. of (ns + (n – 1) d)e^-
f-block, then group no. = 3.
e.g.: Br(Z = 35): $1s^2\ 2s^2\ 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$
Block = p-block (∵ last e^- enters p-subshell)
Period = 4 (∵ largest quantum no. is 4)
Group = 10 + no. of (4s + 4p)e^-
= 10 + (2 + 5)
= 17

(b) If atomic no. (Z) is given
Block: Atomic no. of noble gas (x) = 2, 10, 18, 36, 54, 86, 118
s-block: Z = (x + 1), (x + 2)
p-block: Z = (x), (x–1), (x–2), (x–3), (x–4), (x–5)
f-block: Z = (58 to 71) & (89 to 100)
d-block: Z = rest all
Period: period of next noble gas
Group: 18 + given atomic no.(Z) – atomic no. of next noble gas

			Block	Period	Group
e.g.:	(i)	Z = 12	s	3	18 + 12 – 18 = 2
	(ii)	Z = 9	p	2	18 + 9 – 10 = 17
	(iii)	Z = 21	d	4	18 + 21 – 36 = 3
	(iv)	Z = 33	p	4	18 + 33 – 36 = 15
	(v)	Z = 48	d	5	18 + 48 – 54 = 12
	(vi)	Z = 65	f	6	3
	(vii)	Z = 118	p	7	18 + 118 – 118 = 18
	(viii)	Z = 58	f	6	3
	(ix)	Z = 74	d	6	18 + 74 – 86 = 6
	(x)	Z=81	p	6	18+81-86=13
	(xi)	$[Ar]3d^5 4s^2$	d	4	2+5=7
	(xii)	$[Xe]6s^1$	s	6	1
	(xiii)	$[Kr]4d^{10}5s^2 5p^2$	p	5	10+2+2=14
	(xiv)	$[Xe]4f^{14}5d^1 6s^2$	f	6	3
	(xv)	$[Xe]5d^{10}6s^1$	d	6	11

Q33 **Calculate maximum no. of elements which can be present in the 9th period of periodic table.**

Sol: **Method 1:**

Types of orbitals filled in 9th period:

9s	6g	7f	8d	9p
↓	↓	↓	↓	↓

No. of orbitals = 1 + 9 + 7 + 5 + 3 = 25

No. of elements = no. of e^- = 2 × 25 = 50

Method 2:

No. of elements = $2\left(\frac{n+1}{2}\right)^2$ ($\because$ n is odd)

$$2\left(\frac{9+1}{2}\right)^2 = 2\left(\frac{10}{2}\right)^2 = 50$$

Some important points about Modern Periodic Table

Diagonal Relationship : due to same ionic potential there are three pairs which have similar properties with diagonal element to each other. They are: $(Li \leftrightarrow Mg), (Be \leftrightarrow Al), (B \leftrightarrow Si)$

Bridging elements : Period 3 elements are known as bridging elements as they have valencies similar to subgroup A & subgroup B (according to Mendeleev)

Typical elements : Period 3 elements are known as Typical elements as they represent most of the properties of the elements of their respective periods.

Transuranic : Uranium (Z = 92) is last naturally occurring element, all elements with atomic no $\geq$ 92 are man made and are known as Transuranic elements.

Periodic Table (at room temperature) has :

11 Gaseous Elements : H, N, O, F, Cl, He, Ne, Ar, Kr, Xe, Rn

2 liquid elements : Hg (Metal) + Br (non-Metal)

Effective Nuclear Charge (Z_{eff} or Z*):

In multielectron species, electrons experience lesser attraction than number of protons present in the nucleus.

The decrease in force of attraction is due to inner shell electrons, which is called as screening effect or shielding effect.

Decrement in net force of attraction results in decrement in nuclear charge experienced by outer shell e^-, Net nuclear charge experienced by outer most shell $e^-$is called effective nuclear charge (Z*)

$Z_{eff} = (Z - \sigma)$ where, Z = nuclear charge

Z* = effective nuclear charge

σ = screening constant or shielding constant.

Characteristics of screening constant (σ):

Screening will be caused by same shell e^- or inner shell electrons only.

None of outer shell electrons offer screening to inner shell electrons.

Electrons closer to nucleus offer more shielding, it is known as penetration effect.

Screening offered by e^-: K > L > M > N ...

& screening offered by e^-: s > p > d > f ... (for a particular shell)

Theoretical calculation of screening constant (σ):

SLATER'S RULE:

(1) Write electronic configuration in expanded form.

(2) Rearrange the electronic configuration as follows:
(1s)(2s2p)(3s 3p) (3d) (4s 4p) (4d) (4f) (5s 5p) (5d) (5f) ...

(3)(a) If σ on e^- of ns or np is calculated:

(i) σ for each e^- of ns and np (other than 1) is 0.35

(ii) σ for each e^- of (n −1) shell is 0. 85

(iii) σ for each e^- of (n−2) shell and inner shell is 1

(iv) σ for each e^- in outer shell of ns and np is 0.

(b) If σ on e^- of nd or nf is calculated:

(i) σ for each e^- of nd or nf (other than 1) is 0. 35

(ii) σ for each e^- on LHS of nd or nf (all inner e^-) is 1

(iii) σ for each e^- on RHS of nd or nf (all outer e^-) is 0

e.g.: Find Z* on valence e^- of Nitrogen (Z= 7)

N : $1s^2 2s^2 2p^3$

$\underbrace{(1s^2)}\underbrace{(2s^2 2p^3)}$

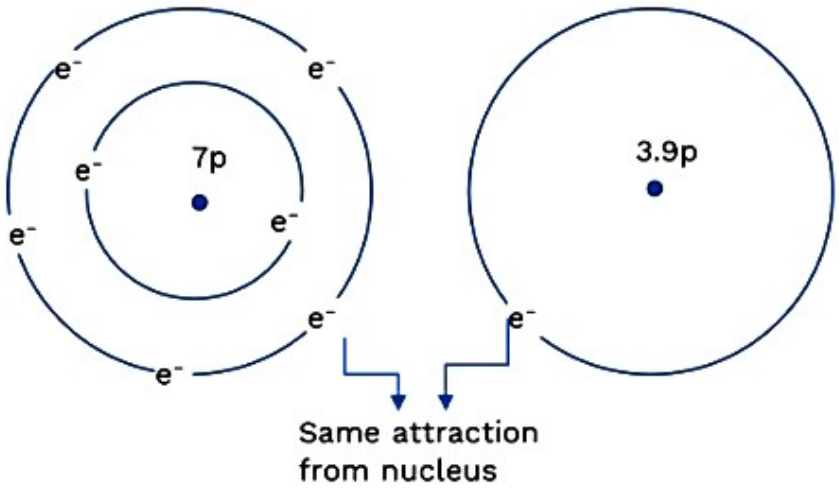
$\sigma = 2 \times 0.85 + 4 \times 0.35$

$\sigma = 1.7 + 1.4$

$\sigma = 3.1$

$Z^* = (z-\sigma) = (7 - 3.1)$

$Z^* = 3.9$



Q34 **How do atomic sizes vary in a group and in a period? Give reasons for the variations.**

Sol: Atomic size decreases along the period due to increase in effective nuclear charge as valence shells are more effectively pulled towards nucleus. On the other hand atomic size increases down the group due to decrease in effective nuclear charge.

Q35 **Find Z* on (i) 4s, (ii) 3d, (iii) 3p, (iv) 3s electron of Fe.**

Sol: (i) 4s e^- of Fe.

$(1s^2)$ $(2s^2 2p^6)$ $(3s^2 3p^6)$ $(3d^6)$ $(4s^2)$

$\sigma = (10 \times 1) + (14 \times 0.85) + (1 \times 0.35)$

$= 10 + 11.9 + 0.35 = 22.25$

$Z^* = 26 - 22.25 = 3.75$

(ii) 3d e^- of Fe.

Fe : $(1s^2)$ $(2s^2\ 2p^6)$ $(3s^2\ 3p^6)$ $(3d^6)$ $(4s^2)$

$\sigma = (18 \times 1) + (5 \times 0.35) + 0$

$\sigma = 18 + 1.75 = 19.75$

$Z^* = 26 - 19.75 = 6.25$

(iii) 3p e^- of Fe

Fe : $(1s^2)$ $(2s^2\ 2p^6)$ $(3s^2 3p^6)$ $(3d^6)$ $(4s^2)$

$\sigma = 2 \times 1 + 8 \times 0.85 + 5 \times 0.35$

$\sigma = 2 + 6.8 + 1.75 = 10.55$

$Z^* = 26 - 10.55 = 15.45$

(iv) 3s e^- of Fe.

Fe : $(1s^2)$ $(2s^2 2p^6)$ $(3s^2 3p^6)$ $(3d^6)$ $(4s^2)$

$\sigma = 2 \times 1 + 8 \times 0.85 + 5 \times 0.35$

$\sigma = 2 + 6.8 + 1.75 = 10.55$

$Z^* = 26 - 10.55 = 15.45$

Q36 **Find Z* on valence electron of Li, Na, K and Mg.**

Sol: Li : $(1s^2)$ $(2s^1)$

$\sigma = 2 \times 0.85 + 0 \times 0.35 = 1.7$

$Z^* = 3 - 1.7 = 1.3$

K : $(1s^2)$ $(2s^2 2p^6)$ $(3s^2 3p^6)$ $(4s^1)$

$\sigma = 10 \times 1 + 8 \times 0.85 + 0 = 16.8$

$Z^* = 19 - 16.8 = 2.2$

Na : $(1s^2)$ $(2s^2\ 2p^6)$ $(3s^1)$

$\sigma = 2 \times 1 + 8 \times 0.85 + 0 \times 0.35 = 8.8$

$Z^* = 11 - 8.8 = 2.2$

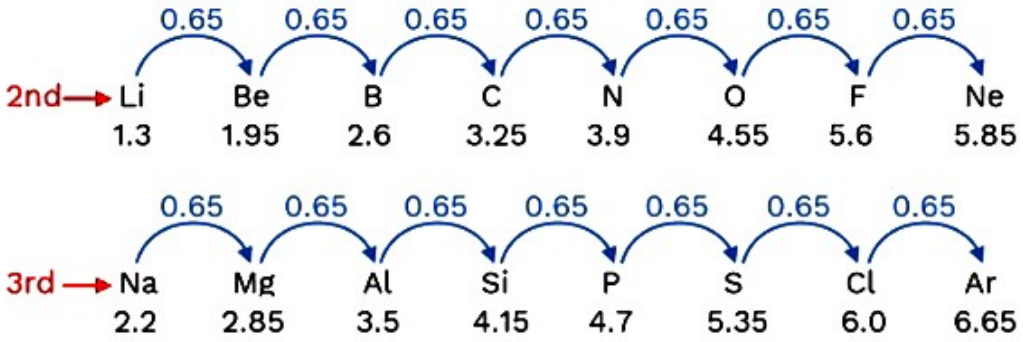
Mg : $(1s^2)$ $(2s^2\ 2p^6)$ $(3s^2)$

$\sigma = 2 \times 1 + 8 \times 0.85 + 1 \times 0.35 = 9.15$

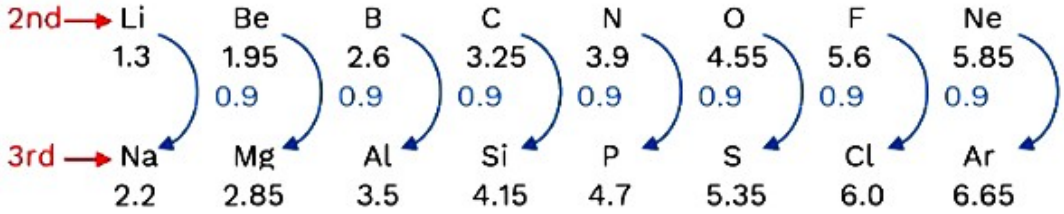
$Z^* = 12 - 9.15 = 2.85$

Observation of Slater:

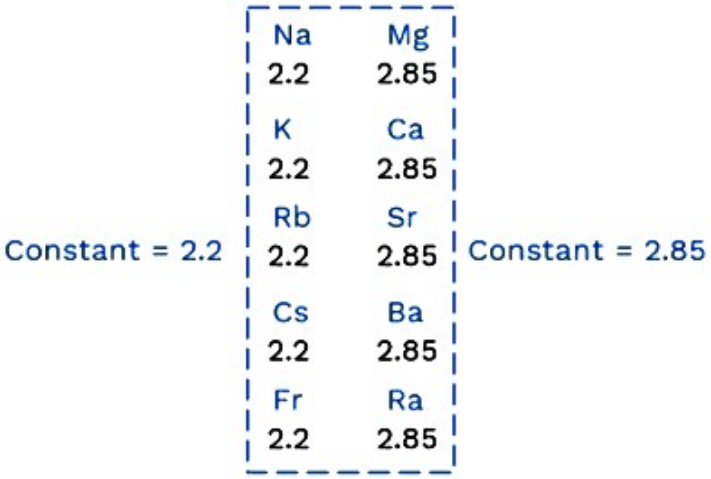
(i) Z* increases by 0.65 in a period in 2nd and 3rd period.



(ii) Z* increases by 0.9 in a group on moving 2nd to 3rd period.



(iii) Z* remains constant in a group on moving 3rd to 7th period in s-block.



(iv) In two electron species like, He, Li^+, Be^{2+}, etc., screening constant of one electron for another is 0.30.

e.g.: In He($1s^2$),

σ = 0.3 (and not 0.35)

Z* = 2−0.3 = 1.7 (and not 1.65)

(v) According to slater, σ and hence Z* on ns and np electron is same.
e.g.: Z* on 3s e^- of Fe = Z* on 3p e^- of Fe = 15.45

Practical observation of effective nuclear charge (Z_{eff})

In a period, on moving from left to right number of proton increases by 1 and effect of proton is 100%, while no. of electron increases by 1 but effect of electron is less than 100%, (since e^- is movable).
So, overall as atomic number (Z) ↑ Z_{eff} ↑.
Also, number of shell remains same and Z_{eff} ↑ so, force of attraction (F_A)↑ (Left to right).

In a group, on moving top to bottom:

(a) In s-block, Z_{eff} increases slightly.
So, increase in shell no.(n) dominates small increase in Z_{eff}.
hence, force of attraction (F_A) decreases down the group.

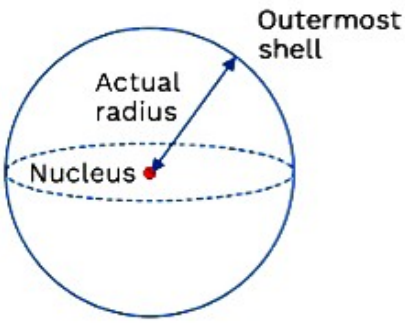
(b) In d-block, Z_{eff} rapidly, due to poor shielding d and f electrons$^-$.
so, large increase in Z_{eff} dominates increase in shell no.(n)
hence force of attraction (F_A) increases down the group.

(c) In p-block, initially Z_{eff} increases rapidly due to less number of p-electrons and large number of d and f electrons.
So, large increase in Z_{eff} dominates increase in shell no.(n)
hence, F_A increases down the group, initially in p-block.

But later from (p^1 to p^6) in p-block, number of p-electrons increases and no. of d and f-electrons remains same, so Z_{eff} increases slowly. so, increase in shell no.(n) dominates small increase in Z_{eff}.
Hence, F_A increases down the group, later in p-block.

Periodic trends in Property of elements:

(I) **Atomic Size :** Theoretically, atomic size is the shortest distance between nucleus and outermost electron of an isolated gaseous atom.



It is difficult to measure the atomic size practically, because:

(i) Existence of isolated atom is difficult.

(ii) Position of moving electron cannot be predicted accurately. (Heisenberg's uncertainty principle).

Therefore, determination of atomic size is done by calculating half of the internuclear distance between two atoms of an element.

Based on the nature of the element and chemical bonds present with in them, different type of radius are defined :

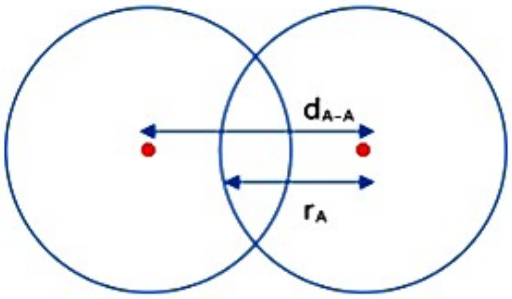
(1) Covalent radius (for non metals)

(2) Metallic radius (for metals)

(3) Vander waal's radius (for Noble gases)

1. Covalent Radius:

Covalent radius is half of internuclear distance between two singly bonded homo atoms.



$$r_A = \frac{d_{A-A}}{2}$$

Three types of covalent radii can be given between two homo atoms

Single bonded covalent radii (SBCR) = half of singly bonded homo atoms

Double bonded covalent radii (DBCR) = half of doubly bonded homo atoms

Triple bonded covalent radii (TBCR) = half of triply bonded homo atoms

e.g. : Distance between c–c > distance between c = c > distance between c ≡ c

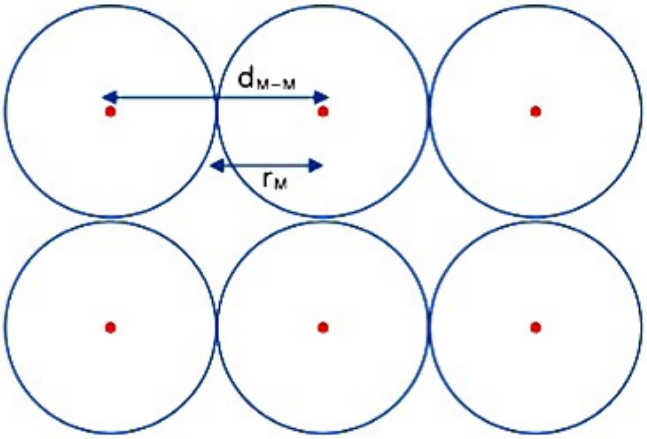
$$d_{c-c} > d_{c=c} > d_{c\equiv c}$$

$$\frac{d_{c-c}}{2} > \frac{d_{c=c}}{2} > \frac{d_{c\equiv c}}{2}$$

SBCR of carbon > DBCR of carbon > TBCR of carbon

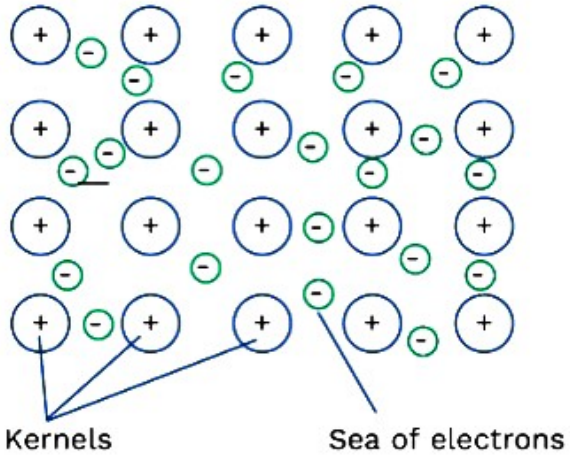
By default, singly bonded covalent radius (SBCR) is covalent radius, because atoms of non-metals are bonded at least by single covalent bond, while they may or may not form double/triple bonds.

2. Metallic Radius:
Half of the internuclear distance between two adjacent atoms of a metal bonded by metallic bond in crystalline lattice structure.



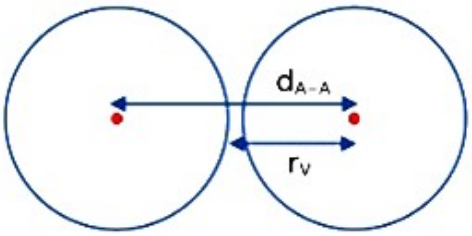
$$r_M = \frac{d_{M-M}}{2}$$

Electron Sea Model (Metallic Bond) :
In early 1900's, paul Drude came up with "Sea of electrons" metallic bonding theory by modelling metals as mixture of kernels (positive nucleus + inner shell electrons) and outer most shell electrons.



An array of positively charged kernels is present in a sea of freely moving electron in gaps of these kernels.
The negative charge of electron sea helps to bind kernels in the array.

Vander waal's Radius : Half of internuclear distance between two adjacent non bonded atoms of an element in a highly compressed state.



$$r_V = \frac{d_{A-A}}{2}$$

In covalent bond, electrons are concentrated between the nucleus while electrons in metallic bonds are movable so attraction in covalent bond is more than that of metallic bond. Also, van der waal forces are weakest, so van der waal radius is largest.

So, for a particular atom:

Covalent radius < Metallic radius ≪ Vander waal radius

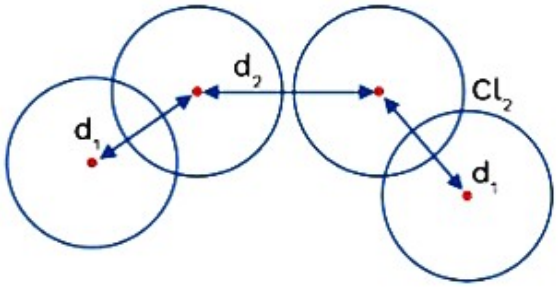
For noble gas, only Vander waal radius is calculated.

For non metals, covalent radius and Vander waal radius, both can be calculated.

e.g.: for chlorine,

Covalent radius of Cl = $\frac{d_1}{2}$

Vanderwaal radius of Cl = $\frac{d_2}{2}$



for metals, Vander waal radius, covalent radius and metallic radius, all three can be calculated.

for lithium :



Element	Radius that may be calculated	Default radius
Metals	Metallic, Covalent, Van der waal	Metallic
Non metals	Covalent, Van der waal	Covalent
Noble gases	Van der waal	Van der waal

Factors affecting Atomic size:

(i) Nuclear charge (Z) : When Z increases, attraction on electron increases and thus size decreases.

(ii) Shielding (σ) : When σ increases, attraction on electron decreases and thus size increases.

(iii) Effective nuclear charge (Z*) : When Z* increases, attraction on electron increases and thus size decreases.

(iv) Shell no. (n) : With increase in shell number size increases.

(v) Crystal Lattice =



Body centered cubic > Face centered cubic

Periodic Variation of atomic size :

(i) In s-block and p-block:

Along the period:

On moving left to right- shell no.(n) remains same, but Z* increases therefore atomic size decreases.

Since Vander waal radius exists for noble gas, they are largest in their respective periods. it is also explained by excessive shielding caused by fully filled orbitals.

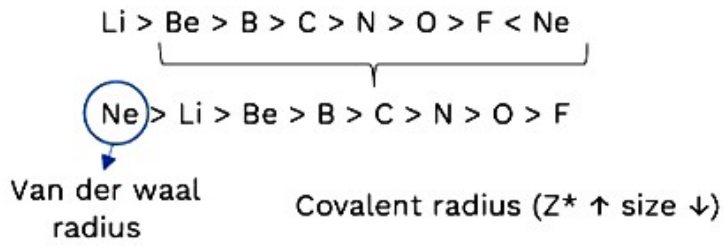
e.g. : size of elements of 2nd period

Li > Be > B > C > N > O > F < Ne

Ne > Li > Be > B > C > N > O > F

Van der waal radius Covalent (Z* ↑ size ↓) radius

e.g.: Order of van der waal radius of element of 2nd period
order of size :



When nothing is mentioned, size of elements is compared on the basis of default radius.
Order of van der waal radius : Li > Be > B > C > N > O > F > Ne → (Z* increases, size decreases)
When radius of elements is compared on same scale then, Z* is only factor in the period and shell no. n is only factor in group.

For period 1, 2, 3 : Size of group 18 > group 1 > group 2	H < He
	Li < Ne
	Na < Ar
For period 4,5,6,7 : size of group 1 > group 18 > group 2...	K > Kr
	Rb > Xe
	Cs > Rn
	Fr > Og

Exception : Pb > Bi (Expected)
Pb < Bi (actual)
due to crystal lattice.

Along the group :
On moving top to bottom shell no.(n) increases so size increases
although the size increases, the rate of increase of size, decreases down the group, due to increase in no. of poor shielding electron.
Exception : In group 13 order of size is :

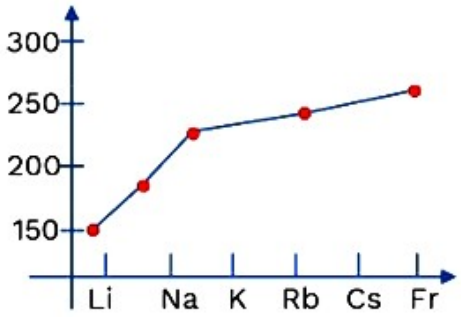
Expected :	B < Al < Ga < In < Tl
Actual :	B < Al > Ga < In < Tl
	B < Ga < Al < In < Tl

due to 10 poor shielding 3d electrons in Ga, Z* increases, increase in Z* dominates over increase in shell no.
In group 14, order of size is : C < Si < Ge < Sn < Pb
(no exception) Shell no. (n) increases so size increases
Top to bottom increment in size decreases.

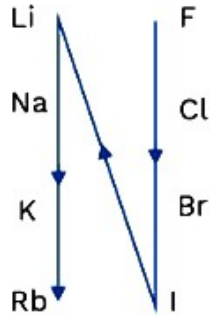
Li Na K Rb Cs Fr

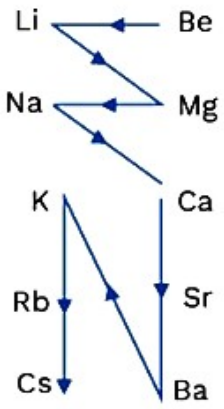
x y z

difference in size x > y > z



Data based orders:



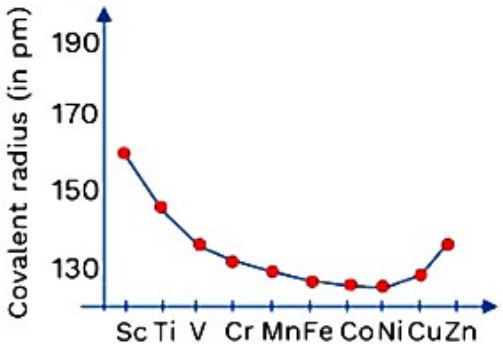


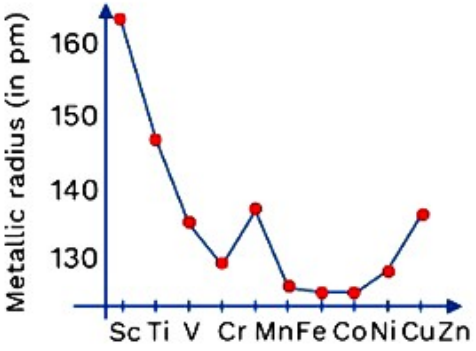
Small → Large

(ii) In d-block:

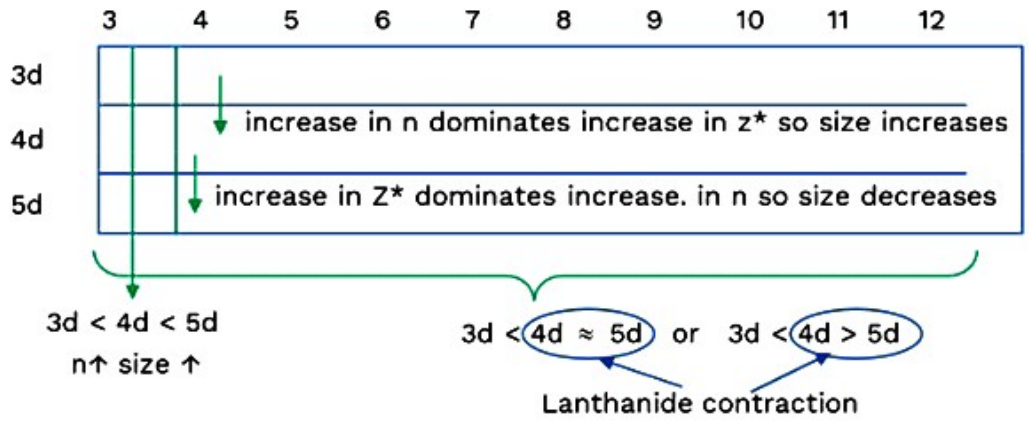
Along the period:

On moving left to right size first decreases, then remain constant in middle and than finally increases.





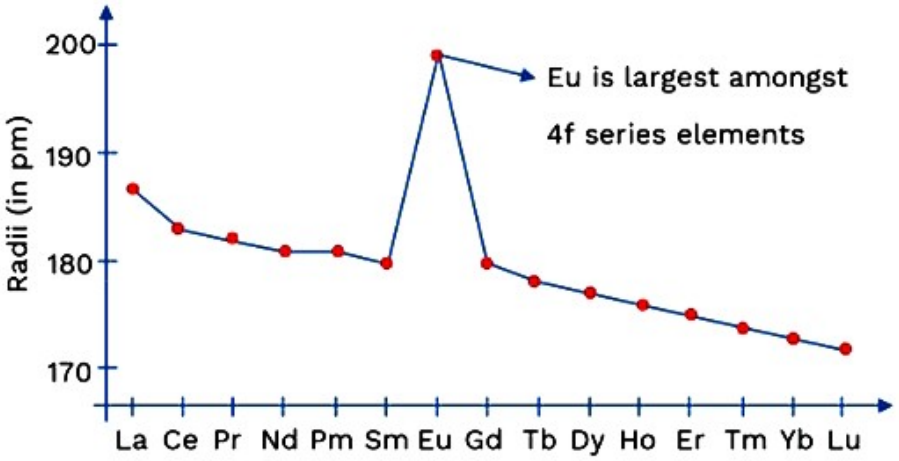
Along the group :



Lanthanide contraction : in d-block on 4d to 5d series after Lanthanum size decreases or remains almost same due to increase in Z* of 14 poor shielding 4f electron, this is known as Lanthanoid contraction.

(iii) In f-block :

on moving left to right size decreases (generally), due to increase in nuclear charge(z).



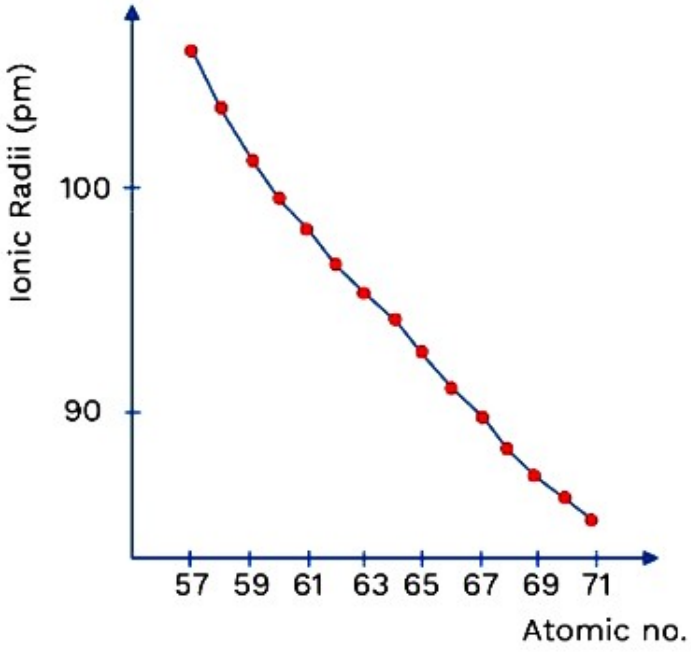
On moving left to right, the decrease in size of lanthanoids is less than decrease in size of actinoids, because number of poor shielding electrons are more in 7th period.

Actinide contraction : decrease in size of elements in 7th period due to poor shielding 5f and 6d electrons.

Actinoid contraction is higher than lanthanoid contraction. due to presence of 4f and 5f electrons in actinoids.

Size of M^{3+} decreases regularly from left to right in 4f series.

$Ce^{3+} > Pr^{3+} > Nd^{3+} > Pm^{3+} > Sm^{3+} > Eu^{3+} > Gd^{3+} > Tb^{3+} > Dy^{3+} > Ho^{3+} > Er^{3+} > Tm^{3+} > yb^{3+} > Lu^{3+}$



Ionic Radius : It is of two types (1) Cationic radius and (2) Anionic Radius

(1) Cationic Radius:

When neutral atom loses electron it forms cation. cation is always smaller than neutral atom. As positive charge on cation increases size of cation decreases.

	$A \longrightarrow$	$A^{+} \longrightarrow$	$A^{+2} \longrightarrow$	A^{+3}
No. of protons	Z	Z	Z	Z
No. of electrons	Z	Z-1	Z-2	Z-3

Z remains same, σ ↓ , z* ↑ , size ↓

(2) Anionic Radius:

When neutral atom gains electron ,it forms anion.

Anion is always larger than neutral atom.

As no. of negative charge on anion increases, size of anion increases.

	$A \longrightarrow$	$A^{-} \longrightarrow$	$A^{-2} \longrightarrow$	A^{-3}
No. of protons	Z	Z	Z	Z
No. of electrons	Z	Z+1	Z+2	Z+3

Z remains same, σ ↑ , z* ↓ , size ↑

(3) Size of isoelectronic species:

It is compared by (Z/e) ratio

As (Z/e) ratio increases ionic radius decreases.

eg :

	Na^+,	Mg^{2+},	F^-,	N^{3-},	Al^{3+}
Z →	11	12	9	7	13
e^- →	10	10	10	10	10
z/e →	1.1	1.2	0.9	0.7	1.3

Z/e order → $Al^{3+} > Mg^{2+} > Na^+ > F^- > N^{3-}$

Ionic radius → $Al^{3+} < Mg^{2+} < Na^+ < F^- < N^{3-}$

Although noble gas are isoelectronic with others, but Z/e can't be used because of different types of radius, ionic radius for ions and van der waal radius for noble gases.

e.g. : $N^{3-} > Ne > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

$P^{3-} > Ar > S^{2-} > Cl^- > K^+ > Ca^{+2} > Sc^{3+}$

Smallest cation = H^+

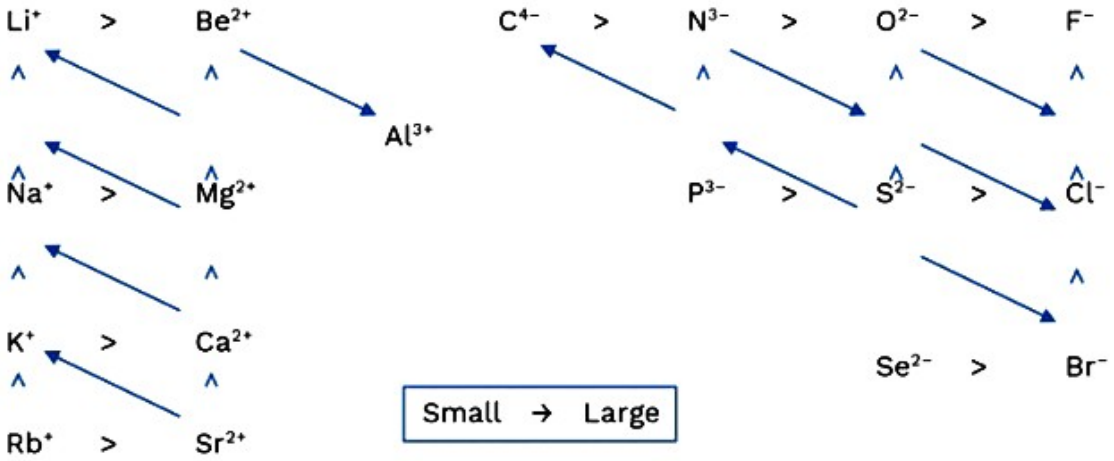
Smallest anion = F^- (not H^-)

	H^-	F^-	Cl^-	Br^-	I^-
$\frac{Z}{e^-}$	$\frac{1}{2}$	$\frac{9}{10}$	$\frac{17}{18}$	$\frac{35}{36}$	$\frac{53}{54}$

Size $H^- > F^- < Cl^- < Br^- < I^-$

$F^- < Cl^- < Br^- < H^- < I^-$

(4) Some ionic radius based on data:



Q37 **Which one is larger van der Waals radius or covalent radius?**

Sol: van der Waals radius.

Q38 **Which one is larger crystal radius (metallic radius) and covalent radius?**

Sol: Metallic radius.

Q39 **Which of the following has smallest ionic radius. Li^+, Na^+, K^+ ?**

Sol: Li^+

Q40 **Arrange the following in order of increasing ionic radius :**
(a) Cl^-, P^{3-}, s^{2-}, F^-;
(b) Al^{3+}, Mg^{2+}, Na^+, O^{2-}, F^-;
(c) Na^+, Mg^{2+}, K^+.

Sol: (a) $F^- < Cl < S^{2-} < P^{3-}$
(b) $Al^{3+} < Mg^{2+} < Na^+ < F^- < O^{2-}$;
(c) $Mg^{2+} < Na^+ < K^+$.

Q41 **Which of the following has largest and smallest ionic radius : I^-,I^+ and I?**

Sol: Largest I^-, Smallest I^+

Q42 **A student reported the atomic (ionic) radii of Cu, Cu^{1+} and Cu^{2+} as 96 pm, 122 pm and 72 pm, respectively. Do you agree with the reported values? Justify with reasons.**

Sol: No; the size of cations is always smaller than its parent atom.

Q43 **Mg^{+2} is smaller than O^{2-} in size, though both have same electronic configuration. Explain.**

Sol: No doubt Mg^{2+} and O^{2-} both has 10 electrons but Mg^{2+} having 12 protons in its nucleus (O^{2-} has 8 protons) exerts higher effective nuclear charge and thus shells are more effectively pulled in Mg^{2+} resulting to smaller size than O^{2-}

Q44 **The tonic radii of S^{2-} and Te^{2-} are 1.84 and 2.21 Å respectively, What may be the ionic radius of Se^{2-} and p^{-3}.**

Sol: ionic radius of Se^{2-} is average of S^{2-} and Te^{2-} i.e. $\frac{1.84 \times 2.21}{2} = 2.02Å$ as 16th gp has O, S, Se, Te, P^{3-} has ionic radius should be larger than S^{2-} because the radii of iso electronics decreases with increase in atomic number.

Q45 **K^+ and F^- have identical radius about 1.34Å. What should be the atomic radius of K and F atoms.**

Sol: Atomic radius of K > ionic radius of K^+
> 1.34 Å
Atomic radius of F < ionic radius of F^-
< 1.34 Å
Cations are always smaller than their parent atom; anions are always larger than their parent atom.

(II) Ionisation energy and Ionisation Enthalpy

Ionisation Energy: It is the amount of energy required to remove most loosely held electron from isolated gaseous atom/ion to form another isolated gaseous ion.

Ionisation Enthalpy (ΔH_{IE}): Change in enthalpy when an electron is removed from outermost shell of isolated gaseous atom/ion.

$A(g) \rightarrow A^+(g) + e^-$ (IE_1 or ΔH_{IE} of A)

By default all ionization energy [IE] values are specified at zero kelvin (OK) hence at any other temperature, change in energy for ionization process is given by ionization enthalpy (ΔH_{IE}).

$$\Delta H_{IE} = IE + \frac{5}{2}RT$$

ΔH_{IE} (at zero kelvin) = IE

Some important points about IE/ΔH_{IE} :

(i) $A(g) \xrightarrow{IE_1 \text{ of A}} A^+(g) \xrightarrow[IE_1 \text{ of } A^+]{IE_2 \text{ of A}} A^{2+}(g) \xrightarrow[IE_1 \text{ of } A^{2+}]{IE_3 \text{ of A}} A^{3+}(g)$

(ii) $A(g) \xrightarrow{IE} A^{3+}(g) + 3e^-$ ($IE = (IE_1 + IE_2 + IE_3)$ of A)

(iii) Ionisation is always endothermic process (energy is required) for neutral atom or cation.

(iv) Successive ionization energy is always increasing

$IE_1 < IE_2 < IE_3 < IE_4 < .....$

	$A \xrightarrow{IE_1} A^+ \xrightarrow{IE_2} A^{2+} \xrightarrow{IE_3} A^{3+}$			
No. of protons	Z	Z	Z	Z
No. of electrons	Z	Z−1	Z−2	Z−3

Z same, σ ↓ , z^* ↑ , IE ↑

(v) It is measured in eV/atom or kcal/mol or kJ/mol

1 eV/atom = 96.4 kJ/mol

4.2 kJ/mol = 1 kcal/mol

Factors Affecting Ionisation Energy/Ionisation Enthalpy:

(i) Effective nuclear charge (Z*) increases, attraction increases, IE increases.

(ii) Screening constant (σ) increases, attraction increases, IE increases.

(iii) Size/shell no.(n) increases, attraction increases, IE decreases.

(iv) Penetration power of subshells increases, attraction increases, IE increases.

Penetration power (P.P.) of s > p > d > f.

eg: IP_1 of Be > B → PP of 2s > 2p

IP_1 of Mg > Al → PP of 3s > 3p

IP, of Zn > Ga → PP of 4s > 4p

(v) Extra stable electronic configuration, half filled (p^3, d^5, f^7) and fully filled (p^6, d^{10}, f^{14}) electronic configuration require slightly higher ionization energy.

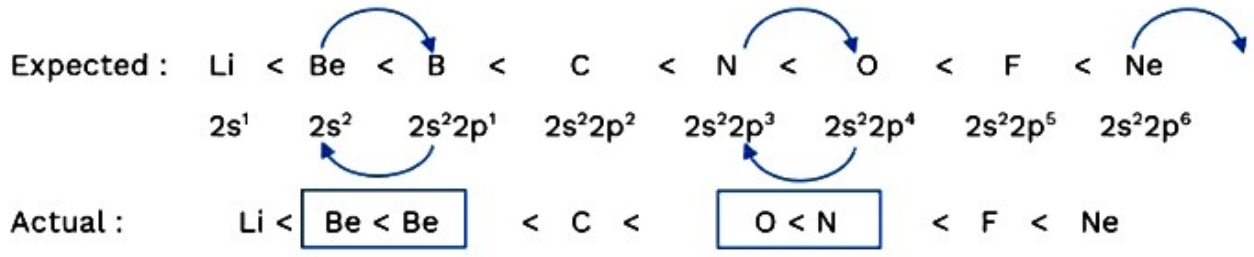
Periodic variations of IE/ΔH_{IE}

Periodic variations of ionization energy (IE), ionization enthalpy (ΔH_{IE}) and ionization potential (IP) are same.

(1) In s-block and p-block:

In a period, on moving left to right IE increases (generally) (extra stable electronic configuration like ns^2 or ns^2np^3 or ns^2np^6 have higher IE than next element).

Example : order of IE_1 of second period elements:



Example : order of IP_1 of s-block and p-block elements:

Period 3 : Na < Al < Mg < Si < S < P < Cl < Ar

period 4 : K < Ga < Ca < Ge < Se < As < Br < Kr

Period 5 : Rb < Sr < In < Sn < Sb < Te < I < Xe

Period 6 : Cs < Ba < Te < Bi < Pb < Po < At < Rn

Comparison of nth ionization energy

for IE_1, configuration of A is compared

for IE_2, configuration of A^+ is compared
for IE_3, configuration of A^{+2} is compared
for IE_n, configuration of A^{+n-1} is compared

Example : $\underset{2p^2}{C} \xrightarrow{IE_1} \underset{2p^1}{C^+} \xrightarrow{IE_2} \underset{2s^2}{C^{+2}} \xrightarrow{IE_3} \underset{2s^1}{C^{+3}}$

$\underset{2p^3}{N} \xrightarrow{IE_1} \underset{2p^2}{N^+} \xrightarrow{IE_2} \underset{2p^1}{N^{+2}} \xrightarrow{IE_3} \underset{2s^2}{N^{+3}}$

$\underset{2p^4}{O} \xrightarrow{IE_1} \underset{2p^3}{O^+} \xrightarrow{IE_2} \underset{2p^2}{O^{+2}} \xrightarrow{IE_3} \underset{2p^1}{O^{+3}}$

$\underset{2p^5}{F} \xrightarrow{IE_1} \underset{2p^4}{F^+} \xrightarrow{IE_2} \underset{2p^3}{F^{+2}} \xrightarrow{IE_3} \underset{2p^4}{F^{+3}}$

Always remove (n –1) e^- in comparison of IE_n.

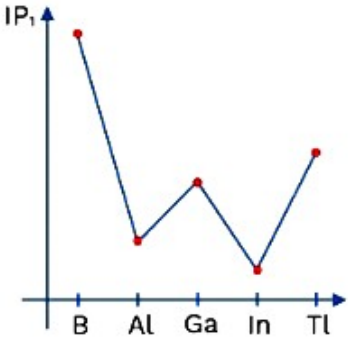
Example : IE_1 $\quad C < N > O < F \Rightarrow C < O < N < F$

IE_2 $\quad C < N < O > F \Rightarrow C < N < F < O$

IE_3 $\quad C > N < O < F \Rightarrow N < C < O < F$

IE_4 $\quad C < N > O < F \Rightarrow C < O < N < F$

In a group, on moving top to bottom IE decreases (generally)
Exceptions : (i) IP_1 of group 13: (more exception)



Expected : B > Al > Ga > In > Tl
Actual : B > Al < Ga > In < Tl
Overall : B > Tl > Ga > Al > In

(ii) IP_1 of group 14 : C > Si > Ge > Sn < Pb
(less exception) C > Si > Ge > Pb > Sn

(iii) IP_1 of group 15 : N > P > As > Sb > Bi
(no exception)

(iv) IP_1 of Sn < Pb

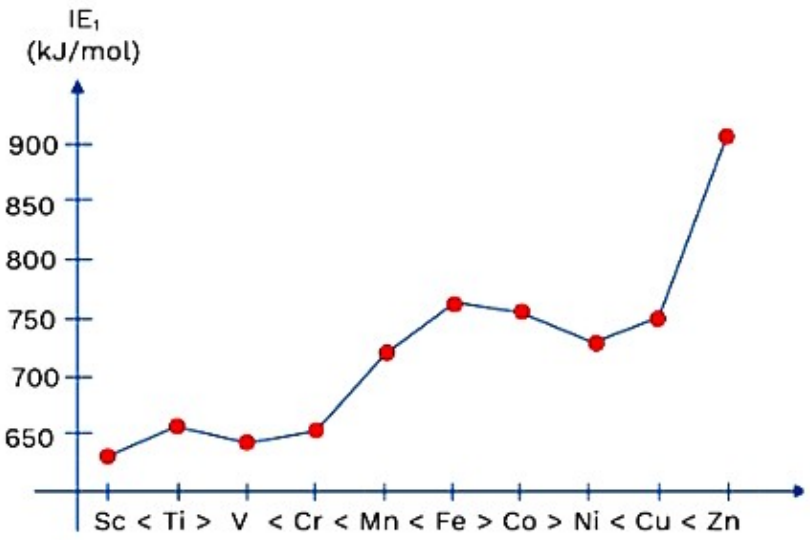
(v) IP_1 of Cs ≈ Fr

(vi) IP_1 of Ba < Ra

(2) In d-block:

In a period, on moving from left to right Z* increases IE increases (Generally) When an electron is removed from (ns), positive charge appears, hence second (ns) electron jumps to (n–1)d. If no. of exchange increases sufficiently, decrement in IE observed.

IE_1 : Sc < Ti > V < Cr < Mn < Fe > Co > Ni < Cu < Zn



IE_2 from left to right increase but Cr and Cu have higher value of IE_2 value than Mn and Zn respectively due to more stable electronic configuration.

IE_2 : Sc < Ti < V < $\boxed{Cr > Mn}$ < Fe < Co < Ni < Cu > Zn

↓ ↓

$Cr^+(d^5)$ $Cu^+(d^{10})$

IE_3 of elements of 3d-series increases form left to right, but IP_3 of Mn > Fe because of extra stable electronic configuration.

IP_3 : Sc < Ti < V < Cr < $\boxed{Mn > Fe}$ < Co < Ni < Cu < Zn

↓

$Mn^{2+}(d^5)$

In a group, on moving top to bottom for :

IP_1 of group 3 : 3d > 4d > 5d > 6d → Sc > Y > La > Ac

IP_1 of group 10 : 5d > 4d > 3d → Pt > Pd > Ni

IP_1 of group 8, 9, 11, 12 : 5d > 3d > 4d → Os > Fe > Ru

Ir > Co > Rh

Au > Cu > Ag

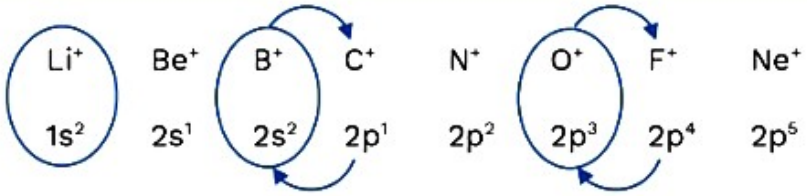
Hg > Zn > Cd

(iii) In f-block :

- On moving left to right size decreases IE increases (generally), due to increase in nuclear charge (Z).

Q46 Give order or IP_2 or 2nd period element?

Sol:



(1) Noble gas configuration → Z* increases IE increases
(2) lower shell

IP_2 : Be < C < B < N < F < O < Ne < Li

Application of $IE/\Delta H_{IE}$

(i) Metallic character $\alpha \frac{1}{IE}$

That is down the group metallic character increases

and left to right in a period metallic character decreases

(ii) Reducing nature $\alpha \frac{1}{IE}$

Reducing nature of group 1 elements (in gas) : Li < Na < K < Rb < Cs

(iii) Elements with lower IE have greater tendency to form cation.

(iv) To determine the No. of valence electron of representative elements:

No. of valence electron = No. of ionization before a sudden jump in successive IE's

Example: Successive IE of an element : 14, 16, 20, 24, 75, 100 no. of valence electron = 4 (sudden jump between IE_3 and IE_4)

(v) Stability of oxidation states of an element:

Depending upon the difference in successive IE stable oxidation state can be predicted.

$[IE_n - IE_{(n-1)}] > 16$ eV/atom : lower oxidation state more stable

$[IE_n - IE_{(n-1)}] < 11$ eV/atom : higher oxidation state more stable

Example: $Al \rightarrow Al^+$, $Al^+ \rightarrow Al^{2+}$] $\Delta IE(IE_2 - IE_1) = 12.8$ eV/atom > 16 eV/atom → lower O.S. stable so Al^+ is more stable than Al^{2+}

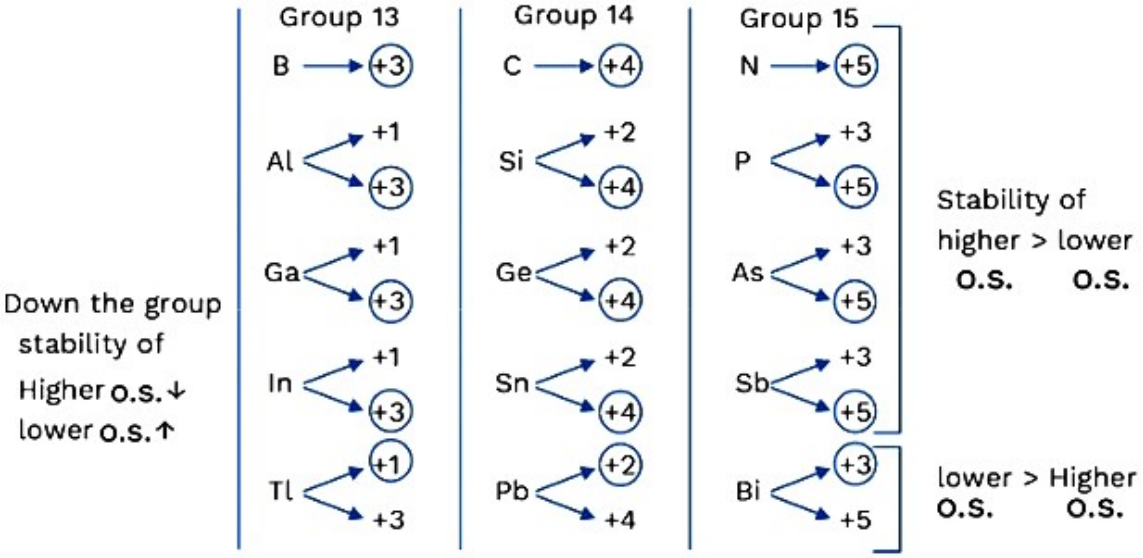
$Al^{+} \rightarrow Al^{2+}$, $Al^{2+} \rightarrow Al^{3+}$] $\Delta IE(IE_3 - IE_2)$ = 6 eV/atom < 11 eV/atom → higher O.S. stable so Al^{3+} is more stable than Al^{2+}

Overall order of stability : $Al^{3+} > Al^{+} > Al^{+2}$

(vi) Inert Pair effect

In p-block, on moving down the group stability of higher oxidation state decreases and stability of lower oxidation state increases.

Because, as we move down the group, no. of poor shielding d and f electrons increases, so attraction on (ns) electron increases which makes (ns)e⁻, inert for bond formation.



Order of stability:

(i) group 13: $Ga^{+1} < In^{+1} < Tl^{+1}$

$Ga^{+3} > In^{+3} > Tl^{3+}$

(ii) group 14: $Ge^{+2} < Sn^{+2} < Pb^{+2}$

$Ge^{+4} > Sn^{+4} > Pb^{+4}$

Down the group stability of Higher O.S.↓

(iii) Group 15 : $As^{+3} < Sb^{+3} < Bi^{+3}$

$As^{+5} < Sb^{+5} > Bi^{+5}$

(iv) $Pb^{+2} > Pb^{+4}$

$Bi^{3+} > Bi^{+5}$

(v) $Sn^{+2} < Sn^{+4}$

Q47 **Among the elements Li, K, Ca, S and Kr, which one is expected to have the lowest first ionization enthalpy and which the highest first ionisation enthalpy?**

Sol: K has lowest first ionisation energy because of larger size. Kr has highest first ionisation energy because of smallest size and ns^2, np^6 configurations.

Q48 **Why does the first ionisation enthalpy increase as we go from left to right across a given period of the Periodic Table?**

Sol: On moving left to right along a period, effective nuclear charge increases. Thus electrons are more tightly held by the nucleus and is therefore their removal is more difficult so the first ionisation enthalpy increases from left to right.

Q49 **Why ionisation enthalpies decrease down the group in the Periodic Table.**

Sol: On moving down a group, valence shells become farther away from the nucleus and therefore electrons are less tightly held with nucleus ionisation enthalpy thus decreases down the group.

Q50 **The first (IE_1) and the second (IE_2) ionization enthalpies (kJ mol^{-1}) of a few elements designated by Roman numerals are shown below:**

Element	IE_1	IE_2
I	2372	5251
II	520	7300
III	900	1760
IV	1680	3380

Which of the above elements is likely to be :

(a) a reactive metal,

(b) a reactive non-metal,

(c) a noble gas,

(d) a metal that forms a stable binary halide of the formulae AX_2 (X = halogen).

Sol: (a) II (low IE_1 and high IE_2).

(b) IV (high IE_1 and IE_2)

(c) I (highest values of IE_1 and IE_2 among the given elements),

(d) III (moderate values of IE_1 and IE_2).

Q51 **Which of the elements Na, Mg, Si and P would have the greatest difference between the first and second ionization enthalpies. Briefly explain your answer.**

Sol: Element which will have stable configuration after removal of one electron will have the maximum difference between IE_1 and IE_2 so the element is Na.

$$Na: 1s^2, 2s^2\ 2p^6, 3s^1 \xrightarrow[\text{from 3s}]{-e} Na^+: 1s^2, 2s^2\ 2p^6 \xrightarrow[\text{from 2p}]{-e} Na^{2+}: 1s^2, 2s^2 2p^5$$

A jump in ionisation enthalpy is noticed only when valence shell changes during successive removal of electron.

Q52 **Select in each pair, the one having lower ionisation energy and explain the reason.**

(a) I and I^-, (b) Br and K, (c) Li and Li^+, (d) Ba and Sr,
(e) O and S, (f) Be and B, (g) N and O.

Sol: (a) I^- (Lower effective nuclear charge).
(b) K (Lower effective nuclear charge),
(c) Li (Lower effective nuclear charge),
(d) Ba (Larger size of the atom),
(e) S (Larger size of the atom),
(f) B (Ellipticity of subshell),
(g) O (Nature of configuration)

Q53 **IE_1 of Be > IE_1 of B, Explain why?**

Sol: Removal of electron from Be takes place from 2s whereas in B it occurs from 2p, 2s orbitals are more closer to nucleus and thus more tightly held and removal of electron requires more energy.

Q54 **I.E. of O < IE_1 of N but IE_2 of O > IE_2 of N. Explain why?**

Sol: $N: 1s^2, 2s^2 2p^3 \xrightarrow[\text{half filled and this more } IE_1]{\text{Removal from 2p}} N^+: 1s^2, 2s^2 2p^2 \rightarrow N^2: 1s^2, 2s^2 2p^1$

$O: 1s^2, 2s^2 2p^4 \rightarrow O^+: 1s^2, 2s^2 2p^3 \xrightarrow[\text{half filled and thus more } IE_2]{\text{Removal from 2p}} O^{2+}: 1s^2, 2s^2 2p^3$

Q55 **Na^+ has higher value of ionisation energy than Ne, though both have same electronic configuration. Explain.**

Sol: No doubt Na^+ and Ne both has 10 electrons but Na^+ having 11 protons in its nucleus (Ne has 10 protons) exerts higher effective nuclear charge and thus removal of electron from Na^+ requires more energy.

Q56 **Why is the second ionisation energy of lithium much greater than that of beryllium?**

Sol: The electronic configuration of Li^+ is $1s^2$ and that of Be^+ is $1s^2,2s^1$. It requires much more energy to remove a 1s core electron close to the nucleus of Li^+ than a 2s valence electron further from the nucleus of Be^{1+}

(iii) Electron Affinity / Electron Gain Enthalpy (ΔH_{eg}):

Electron Affinity (EA) : It is the amount of energy released when an electron is added to an isolated gaseous atom or ion.

Electron gain Enthalpy (ΔH_{eg}) : Change in enthalpy when an electron is added to an isolated gaseous atom or ion.

By default all electron affinity values are given at zero kelvin (0 K), but at other temperature electron gain enthalpy is given by

$$\Delta H_{eg} = -EA - \frac{5}{2}RT$$

ΔH_{eg}(at zero kelvin) = – EA

$$A(g) \xrightarrow[\text{may be +ve or -ve}]{EA_1 \text{ of } A} A^-(g) \xrightarrow[\text{always +ve}]{EA_2 \text{ of } A} A^{-2}(g)$$

ΔH_{eg1} may be -ve

ΔH_{eg2} always -ve

or +ve for neutral atom for neutral atom

Example : in the process,

$$A(g) + e^- \xrightarrow{EA} A^-(g)$$

If energy released is 20 eV/atom then EA = 20 eV/atom

and ΔH_{eg} = –20 eV/atom

$$A^-(g) + e^- \xrightarrow{EA} A^{2-}(g)$$

If energy absorbed is 30 eV/atom then EA = –30 eV/atom

and ΔH_{eg} = 30 eV/atom

Some important points about EA/ΔH_{eg} :

(i) $$A(g) \xrightarrow{EA_1 \text{ of } A} A^-(g) \xrightarrow[EA_1 \text{ of } A^-]{EA_2 \text{ of } A} A^{2-}(g) \xrightarrow[EA_2 \text{ of } A^- \; EA_1 \text{ of } A^{2-}]{EA_3 \text{ of } A} A^{3-}(g)$$

(ii) $A(g) + 3e^- \xrightarrow{EA} A^{3-}(g)$ [EA = (EA_1 + EA_2 + EA_3) of A]

(iii) Magnitude of successive EA of neutral atom increases

$|EA_1| < |EA_2| < |EA_3| < ...$

(iv) Generally ,first electron addition of an isolated gaseous atom is an exothermic process.
(except for stable electronic configurations)

(v) Second Electron addition of an isolated gaseous atom is always endothermic process due to interelectronic repulsion

Example: $O(g) + e^- \rightarrow O^-(g)$ $\quad \Delta H_{eg1} = -141$ kJ/mol (exothermic)

$O^-(g) + e^- \rightarrow O^{-2}(g)$ $\quad \Delta H_{eg2} = +744$ kJ/mol (endothermic)

$O(g) + 2e^- \rightarrow O^{-2}(g)$ $\quad \Delta H_{eg} = +603$ kJ/mol (endothermic)

(vi) Formation of polynegative anions (O^{2-}, N^{3-} etc) is always endothermic

$$\underbrace{A(g) \xrightarrow{EA_1 \text{ of } A} A^-(g) \xrightarrow{EA_2 \text{ of } A} A^{2-}(g)}_{EA_1+EA_2=\text{always -ve}}$$

Relation between Electron affinity and ionization energy :

(i) Electron affinity of neutral atom = ionization energy of its anion

$$A(g) + e^- \underset{IA_1}{\overset{EA_1}{\rightleftharpoons}} A^-(g)$$

EA_1 of A = IE_1 of A^- ...(1)

(ii) Ionisation energy of neutral atom = electron affinity of its cation

$$A(g) \underset{EA_1}{\overset{IA_1}{\rightleftharpoons}} A^+(g) + e^-$$

IE_1 of A = EA_1 of A^+ ...(2)

(iii) Ionisation energy of neutral atom is always greater than electron affinity of neutral atom

$$A^- \underset{EA_1 \text{ of } A^-}{\overset{EA_1 \text{ of } A}{\rightleftharpoons}} A \xrightarrow[IE_2 \text{ of } A^-]{IE_1 \text{ of } A} A^+$$

$\because$ successive IE is always increasing

So IE_1 of A^- < IE_2 of A^-

EA_1 of A < IE_1 of A ...(3)

	EA	ΔH_{eg}	IE	ΔH_{IE}
If energy is released	+ve	–ve	–ve	–ve
If energy is absorbed	–ve	+ve	+ve	+ve

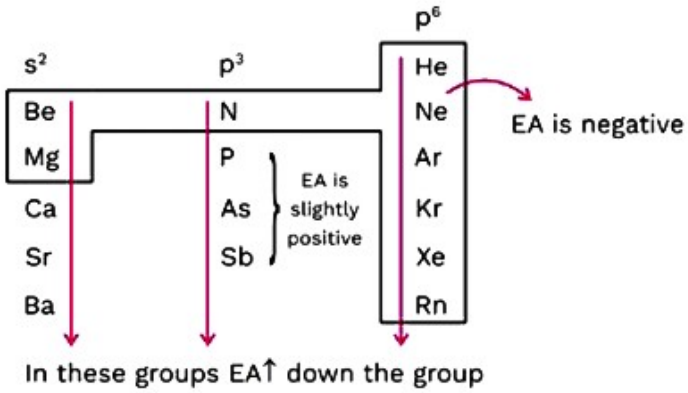
Factors affecting electron affinity :

(i) Effective nuclear charge (Z*) increases, attraction decreases, EA decreases

(ii) Screening constant (σ) increases, attraction decreases, EA decreases

(iii) Shell no. (n) increases, attraction decreases, EA decreases

(iv) Electron affinity of elements having extra stable electronic configuration (half filled and fully filled) is very less or negative. For such elements electron gain enthalpy (ΔH_{eg}) will be positive.



Stability of half filled < fully filled ($ns^2 < ns^2\ np^6$)

So, EA of half filled > fully filled ($ns^2 > ns^2\ np^6$)

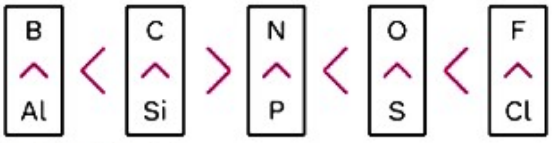
e.g., $\left.\begin{matrix} Ne < Be < N \\ Ar < Mg < P \end{matrix}\right\}$ EA of $ns^2\ np^6 < ns^2 < ns^2\ np^3$

(v) Penetration power increases, EA increases

$$\left.\begin{matrix} EA_1 \text{ of } Li > B \\ EA_1 \text{ of } Na > Al \\ EA_1 \text{ of } K > Ga \end{matrix}\right\} \text{(penetration of ns > np)}$$

e.g.,

(vi) Electron density of 2nd period > 3rd period elements. So, addition of electrons to 2nd period elements is difficult, hence EA of 2nd period element is less than 3rd period element (in p-block).



e.g., EA_1 of O < S < F < Cl

EA_1 of N < P < O < S

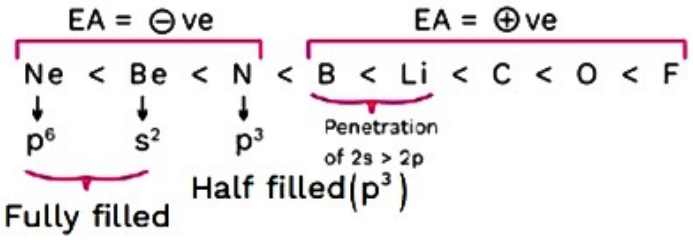
Periodic variations of electron affinity :

In s-block and p-block :

In a period :

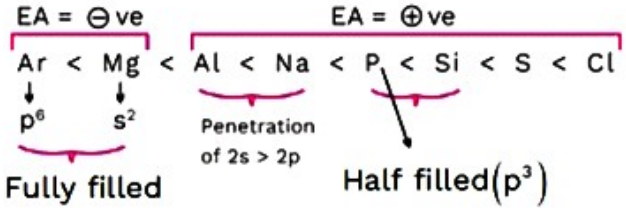
- EA of 2nd period elements :





- EA of 3rd period elements :





In a group :

- EA of group 17 : $\boxed{F < Cl} > Br > I \quad \Rightarrow \quad \boxed{Cl > F} > Br > I$
- EA of group 16 : $\boxed{O < S} > Se > Te \quad \Rightarrow \quad S > Se > Te > \boxed{O}$

Illustration :

Compare IE of F, F^-, Cl, Cl^-.

Solution :

(1) IE of F > Cl ⇒ Down the group IE↓

(2) IE of F > F^- ⇒ Successive IE is higher

(3) IE of Cl > Cl^- ⇒ Successive IE is higher

(4) IE of F^- > Cl^- ⇒ EA of F < Cl

From (1) to (4) ⇒ IE of F > Cl > Cl^- > F^-

Q57 **Electron affinity depends on:**

(A) Atomic size **(B) Nuclear charge**

(C) Atomic number **(D) Atomic size and nuclear charge both**

Sol: **D**

Electron affinity depends on both

Top to bottom atomic size Cl, Br, I, and left to right $N \propto O < F$ Zeffective

Q58 **The correct order of electron affinity of B, C, N, O is**

(A) O > C > N > B **(B) B > N > C > D**

(C) O > C > B > N **(D) O > B > C > D**

Sol: **C**

B, C, N, O

Nitrogen → electron affinity is negative

N < B < C < O

Q59 **Which one has maximum electron affinity.**

(A) N **(B) Be** **(C) B** **(D) Cl**

Sol: **D**

Chlorine has maximum electron affinity.

Q60 **The electron affinity for the Inert gas is:**

(A) Zero **(B) High** **(C) Negative** **(D) Positive**

Sol: **B**

Lowest electron affinity Neon So, electron affinity for the Inert gas element is Negative.

Q61 **Which of the following species has the highest elected affinity.**

(A) F **(B) O** **(C) O^-** **(D) Na^+**

Sol: **D**

Sodium is a positive charge

Oxygen is a Negative charge

$O^- \xrightarrow{e^-}$ Repulsion

F,O → Neutral

Sodium Na^+ is cation

$Na^+ \xrightarrow{e^-}$ attraction ↑

then, electron affinity of sodium is highest

Q62 **Which of then is correct order of Ionization energy.**

(A) $F < F^- < Cl < Cl^-$ **(B)** $F^- < Cl^- < Cl < F$

(C) $Cl^- < Cl < F^- < F$ **(D)** $Cl^- < F^- < Cl < F$

Sol: **B**

I.E = F > Cl

I.E = F^- < F because successive I.E is always increases

I.E = Cl^- < Cl

Q63 **Which of the following pairs of elements would have a more negative electron gain enthalpy? Explain.**

(a) N or O, **(b) F or Cl.**

Sol: (a) O : In N, half filled nature being more stable opposes the addition of electron.

(b) Cl : In F, smaller size of F shows more charge density and thus electron-electron repulsion predominates.

Q64 **Which group members possess II EA to be zero and why?**

Sol: Halogens; after addition of one electron they acquire nearest noble gas configuration $ns^2\ np^6$ and thus have no scope for addition of second electron.

$$\underset{ns^2np^5}{X} + e \rightarrow \underset{ns^2np^6}{X^-}$$

Q65 **Which of the following elements will have**
(i) greatest electron affinity; (ii) lowest electron affinity :
(a) $1s^2, 2s^2 2p^3$; (b) $1s^2, 2s^2 2p^4$; (c) $1s^2, 2s^2 2p^5$; (d) $1s^2, 2s^1$.

Sol: (i) $1s^2,2s^2 2p^5$ has greatest electron affinity
(ii) $1s^2, 2s^2 2p^3$ has lowest electron affinity.

Q66 **Which of the following elements possesses zero electron affinity and why?**
(a) N, (b) O, (c) Ne, (d) Li .

Sol: Neon. It has stable electronic configuration $2s^2 2p^6$ and thus has absolutely no tendency to add an extra electron in higher energy level.

Q67 **Why is second EA for halogens zero?**

Sol: After addition of electron in halogen atom formation of halide ion takes place which has full filled configuration (ns^2np^6) and thus possesses no tendency to gain one more electron.

Q68 **Fluorine is more powerful oxidant than chlorine, Explain.**

Sol: On account of smaller size, F^- possesses higher heat of hydration and thus ΔH_{net} for F_2 is highly exothermic.

$$\frac{1}{2}F_2 \rightarrow F;\ \Delta H_1 = \frac{1}{2}\text{Heat of dissociation}$$

$$F + e \rightarrow F^-;\ \Delta H_2 = -EA_1$$

$$F^- + aq \rightarrow F^-(aq.);\ \Delta H_3 = -\text{Heat of hydration}$$

$$\therefore\ \frac{1}{2}F_2 + e + aq \rightarrow F^-(aq.);\ \Delta H_1 + \Delta H_2 + \Delta H_3$$

Q69 **Electron affinity of chlorine is more than fluorine. Explain.**

Sol: The orbitals that hold the outermost electrons of the halogens are increasingly spread out as we proceed from fluorine to iodine. Due to smallest size of the outermost shell in fluorine, the addition of an extra electron results in strong electron-electron repulsion. As a result, attractive interaction between the incoming electron and nucleus of fluorine atom is considerably counter balanced. This results in low electron affinity for fluorine.

Q70 **Addition of an electron to $Na_{(g)}$ is slightly exothermic process, whereas addition of electron to $Mg_{(g)}$ is strongly endothermic. Explain.**

Sol: In $Mg_{(g)}$, the two outer shell electrons are paired in 3s orbital. An added electron would need to go into a 3p orbital; it would be effectively shielded from the nucleus. So electron-electron repulsion would dominate over the electron-nuclear attraction.
In Na^+ the added electron can pair with the one 3s electron already there. It sees some nuclear charge because it is not completely shielded by the outer most 3s electron.

Q71 **What is the value of electron affinity of Na^+ if IE_1 of Na = 5.1 eV**

Sol: $Na \rightarrow Na^+_{(g)} + e$ $\quad IE_1 = 5.1$ eV

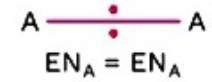
$Na^+_{(g)} + e \rightarrow Na$ $\quad \therefore EA_1 = -IE_1 = -5.1$

Q72 **Which in each pair has higher electron affinity?**
(a) F^- or Cl^- **(b) O or O** **(c) Na^+ or F** **(d) F or F^-**

Sol: (a) F : EA_1 for Cl^- is zero
(b) O^- : EA_2 of O > EA_1 of O and more positive
(c) Na^+
(d) F : EA_2 for F = 0

(III) Electronegativity :

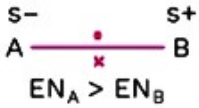
- The tendency of an element to attract shared pair of e^- in a covalent bond is called electronegativity.
- It is not a measurable quantity. However, a relative numerical value is given on various scales.



$EN_A = EN_A$

$\Delta EN = 0$

Non-polar bond



$EN_A > EN_B$

$\Delta EN \neq 0$

Polar bond

- Electronegativity of some elements on Pauling's scale are :

						H 2.1
Li 1.0	Be 1.5	B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2	Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8		Practically EN of : P > H N > Cl S > C				Br 2.8
Rb 0.8						I 2.6
Cs 0.7						

Factors affecting electronegativity :

(i) Nuclear charge (Z) increases, attraction increases, EN increases

(ii) Screening constant (σ) increases, attraction decreases, EN decreases

(iii) Effective nuclear charge (Z*) increases, attraction increases, EN increases

(iv) Shell no. (n) increases, attraction decreases, EN increases

(v) Oxidation state increases, attraction increases, EN increases
e.g., $Fe^{3+} > Fe^{2+}$

(vi) %S – character increases, attraction increases, EN increases
e.g., EN of sp^3 hybridised C < sp^2 hybridised C < sp hybridised C

(vii) EN of surrounding atom increases, EN of central atom increases,
e.g., EN of C in $CH_4 < CH_3I < CH_3Cl < CH_3F$ (EN of F > Cl > I > H)

(viii) No. of mare electronegative surrounding atom increases, EN of central atom increases
e.g., EN of C in $CH_4 < CH_3F < CH_2F_2 < CHF_3$

Calculation of EN of elements :

(i) Pauling's method (on Pauling's scale) : Based on bond energies

$$|X_A - X_B| = 0.208\sqrt{E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}}} = 0.208\sqrt{\Delta}$$

Where, E_{A-A} = Bond energy of pure covalent bond of A – A in kcal/mole
E_{B-B} = Bond energy of pure covalent bond of B – B in kcal/mole
E_{A-B} = Bond energy of pure covalent bond of A – B in kcal/mole
X_A and X_B are EN of A and B respectively
Δ = ionic resonance energy.

(ii) Mulliken's method (on Mulliken's scale) :

EN of element A on Mulliken scale $= \dfrac{\text{IE of A} + \text{EA of A}}{2}$

$$X_M = \frac{IE + EA}{2}$$ IE and EA are in eV/atom

Numerically EN of element on Mulliken scale is larger value than EN of that element on Pauling scale.

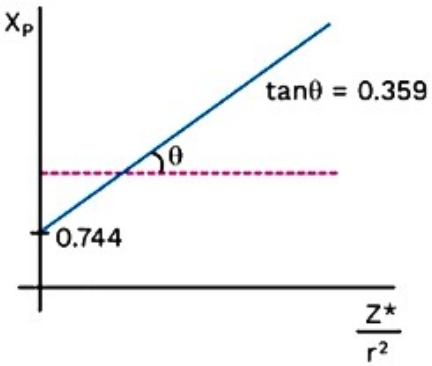
$$X_P \approx \frac{X_M}{2.8} = \frac{IE + EA}{5.6}$$

Where, X_P = EN of A on Pauling scale
X_M = EN of A on Mulliken scale

(iii) Allred-Roschov's Method (On Pauling's scale) :

$$X_p = 0.359\frac{Z^*}{r^2} + 0.744$$

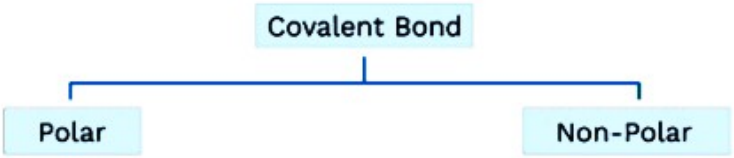
Where, X_P = EN of A on Pauling scale
Z^* = Effective Nuclear charge
r = Covalent radius of A (in Å)



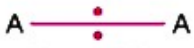
Applications of Electronegativity :

(i) Metallic and Non-metallic nature :
Generally metals have low EN and non-metals have high EN.
So, non-metallic nature $\propto$ EN

(ii) Polarity of bond $\propto$ EN



e^- cloud is unsymmetrically distributed between bonded atoms



$EN_A = EN_A$

$\Delta EN = 0$

Non-polar bond

e^- cloud is symmetrically distributed between bonded atoms



$EN_A > EN_B$

$\Delta EN \neq 0$

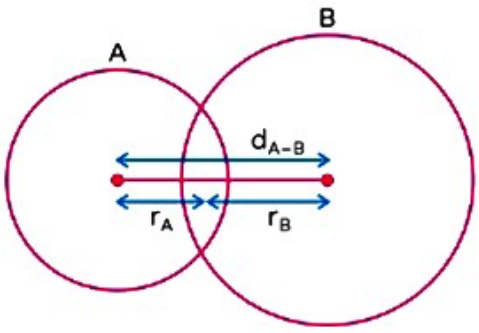
Polar bond

(iii) Hanney-Smith equation :
% Ionic character $16\,|X_A - X_B| + 3.5\,(X_A - X_B)^2$
As $(X_A - X_B)$ increases % ionic character in covalent bond increases
If $\Delta EN = 2.1$ then ionic character = covalent character
If $\Delta EN > 2.1$ then AB is ionic bond
If $\Delta EN < 2.1$ then AB is covalent

(iv) Bond Energy $\propto \Delta EN$
As ΔEN increases, ionic character increases, Bond energy increases

(v) Bond length (Shoemaker and Stevenson Equation) :

Theoretically : For Heteroatomic molecule



$d_{A-B} = r_A + r_B$

Also, $r_A \neq r_B \neq \frac{d_{A-B}}{2}$

- Practically, due to ΔEN, partial charge generates on atoms and extra attraction is observed.
  So, observed d_{A-B} < theoretical d_{A-B}
- Shoemaker and Stevenson Equation

$$d_{A-B} = r_A + r_B - 0.09\left|x_A - x_B\right|$$

Where, d_{A-B} = Internuclear distance between A and B or bond length of A–B
r_A = covalent radius of A
r_B = covalent radius of B

- Bond length $\alpha \frac{1}{\Delta EN}$

(vi) Nomenclature of inorganic compounds :

- While writing names of covalent compounds, names of atoms with lower EN are written first followed by names of atoms with higher EN
- e.g., : $Cl_2O \rightarrow$ dichloro-oxide ✓
  oxo-dichloride ×
- While writing molecular formula of compound, atoms with lower EN are written on left and atoms with higher EN are written on right
- e.g., : Oxygen difluoride → F_2O ×
  OF_2 ✓

(vii) Nature of Hydride (H – A) :

$$\underset{\text{acid}}{HA} + H_2O \longrightarrow H^+ + \underset{\text{conjugate base}}{A^-}$$

- Stability of conjugate base (A^-) increases, tendency to loose H^+ increases, hence, acidic nature of acid (HA) increases.
- In a group
  Stability of A^- ∝ size of A ∝ Acidic nature of HA
- In a period
  Stability of A^- ∝ EN of A ∝ Acidic nature of HA
  e.g., acidic nature of NH_3, H_2O, HF

$$NH_3 \longrightarrow NH_2^- + H^+$$

$$H_2O \longrightarrow OH^- + H^+$$

$$HF \longrightarrow F^- + H^+$$

In period, on moving left to right size remains almost same due to same no of shell, but EN of atom increases so, stability of (–)ve charge increases

hence acidic nature increases.

Stability of $NH_2^- < OH^- < F^-$

Acidic nature of $NH_3 < H_2O < HF$

e.g., acidic nature Of HF, HCl, HBr, HI
down the group, size increases, dispersion of (–)ve charge increases, stability of conjugate base increases, acidic nature of acid increases.

$HF \longrightarrow H^+ + F^-$

$HCl \longrightarrow H^+ + Cl^-$

$HBr \longrightarrow H^+ + Br^-$

$HI \longrightarrow H^+ + I^-$

Stability : $F^- < Cl^- < Br^- < I^-$

Acidic nature : HF < HCl < HBr < HI

(viii) Oxidation number / oxidation state :

- It is a number assigned to an element according to ΔEN
- It is equal charge on an atom, if the bond is ionic
- If EN of A > EN of B

	oxidation state of A	oxidation state of B
A – B	–1	+1
A = B	–2	+2
A ≡ B	–3	+3

e.g., $\overset{+1}{H}-\overset{-1}{Cl}$

$\because$ $EN_H < EN_{Cl}$

$\therefore$ oxidation state of H = +1
oxidation state of Cl = –1

e.g., $\overset{-2}{O} = \overset{+2}{\overset{+2}{C}} = \overset{-2}{O}$

$EN_O > EN_C$

$\therefore$ oxidation state of O = –2
oxidation state of C = +2 +2 = +4

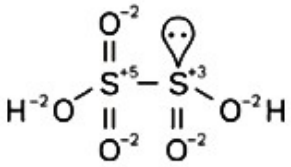
- For ionic compounds oxidation state = Actual charge
  e.g., In NaCl, oxidation state of Na = +1, oxidation state of Cl = –1
- Average oxidation state = It is calculated by using formula.
  Sum of oxidation state of all atoms = net charge
  (O.S. of H = +1) (O.S. of O = –2)

e.g., : $H_3PO_4 \Rightarrow 3(+1) + (x) + 4(-2) = 0 \Rightarrow x = +5$

e.g., : $H_2S_2O_6 \Rightarrow 2(+1) + 2(x) + 5(-2) = 0 \Rightarrow x = +4$

- Absolute O.S. = It is calculated by using structure

e.g., :



(ix) Nature of oxides (A – O) :

- Binary compounds of oxygen, in which oxidation state of oxygen is (–2)

  e.g., Na_2O, K_2O, MgO, CO_2, ZnO are oxides

  F_2O is not an oxide, as O.S. of 'O' = +2

  K_2O_2, H_2O_2, BaO_2 are not oxides, as O.S. of 'O' = –1

  NaO_2, LiO_2, KO_2 are not oxides, as O.S. of 'O' = –1/2
- Metallic oxides, generally gives base in water so they are basic oxides.

$$\underset{\text{Basic oxide}}{Na_2O} + H_2O \longrightarrow \underset{\text{Base}}{NaOH}$$

- Non-metallic oxide gives acids in water so they are acidic oxides.

$$\underset{\text{Acidic oxide}}{Cl_2O} + H_2O \longrightarrow \underset{\text{Acid}}{HOCl}$$

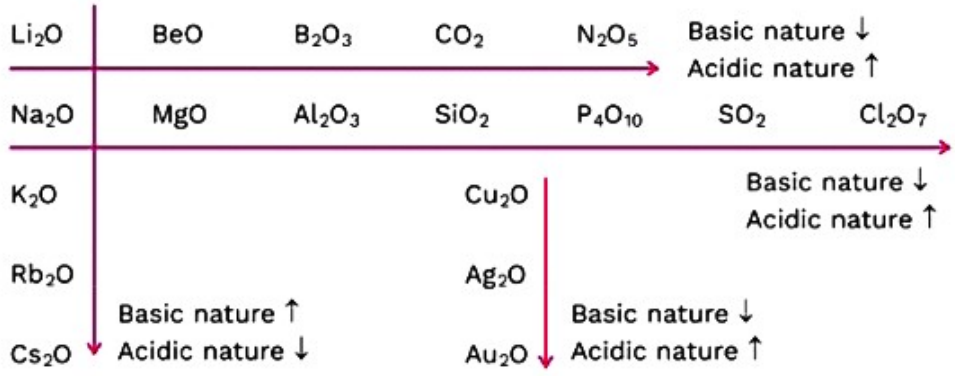
- Acidic nature of oxides $\propto$ EN
  $\propto$ Oxidation state of A
  $\propto$ Non-metallic nature of A
  $\propto \dfrac{1}{\text{metallic nature of A}}$
- Left to right, EN increases, Basic nature decreases, Acidic nature increases
- Top to bottom,

  For s-block, EN decreases, Basic nature increases, Acidic nature decreases,

  For p-block, EN decreases, Basic nature increases, Acidic nature decreases,

  For d-block, EN increases, Basic nature↓ Acidic nature increases

  e.g. :



Example: Acidic nature of :

(i) $\overset{+6}{S}O_3 > \overset{+4}{S}O_2$ O.S↑ Acidic Nature↑

(ii) $\overset{+5}{N_2}O_5 > \overset{+3}{N_2}O_3$ O.S↑ Acidic Nature↑

(iii) $\overset{+5}{Sb_2}O_5 > \overset{+3}{Sb_2}O_3$ O.S↑ Acidic Nature↑

(iv) $\overset{+4}{Si}O_2 < \overset{+5}{P_2}O_5 < \overset{+7}{Cl_2}O_7$ O.S↑ Acidic Nature↑

(v) $SO_3 > SeO_3 > TeO_3$ Non-metallic character↑

(vi) $N_2O_3 > P_2O_3 > Sb_2O_3$ Non-metallic character↑

Q73 **Which two elements would you expect to combine in the most violent ions?**

Sol: Cs and F : Cs is most electropositive and F is most electronegative.

Q74 **Which is more electronegative in each pair?**
(a) Ne or F **(b) F or Cl.**

Sol: (a) F (b) F

Types of Oxides :

(i) Acidic oxides : non-metallic oxides are generally acidic oxides which reacts with base to give salt & water. Aqueous solution is acidic.

$CO_2 + H_2O \longrightarrow H_2CO_3\,(2H^+ + CO_3^{2-})$

$B_2O_3 + H_2O \longrightarrow H_3BO_3\,(3H^+ + BO_3^{3-})$

e.g. : SO_2, P_2O_5, N_2O_3, N_2O_5, CO_2, SiO_2, TeO_3, SeO_3 etc.

(ii) Basic oxides : metallic oxides are generally basic oxides which react with acids to give salt & water. Aqueous solution is basic.

$CaO + H_2O \longrightarrow Ca(OH)_2\ (Ca^{2+} + 2OH^-)$

e.g. : Na_2O, K_2O, MgO, CaO, Cu_2O, Al_2O_3 etc.

(iii) Amphoteric oxides : oxides which react with acid as well as base to give salt and water aqueous solution is basic as well as acidic.

$Al_2O_3 + H_2O \longrightarrow Al(OH)_3$

$Al(OH)_3 + HCl \longrightarrow AlCl_3 + H_2O$

$Al(OH)_3 + NaOH \longrightarrow Na[Al(OH)_4]$

e.g. :

Zn	Be	Al	Ga	Sn	Cr^{+3}	V^{+5}	Sb^{+3}	As^{+3}	Pb
ZnO	BeO	Al_2O_3	Ga_2O_3	SnO SnO_2	Cr_2O_3	V_2O_5	Sb_2O_3	As_2O_3	PbO PbO_2

Metal oxides with (O.S = +4) $\Rightarrow$ Amphoteric [VO_2 (Basic)]

(iv) Neutral oxides : oxides neither reacts with acid nor with base.

e.g. : CO, NO, NO_2

Metallic oxides
- +1, +2, +3 — Basic — except amphoteric
- +4 — Amphoteric — except VO_2
- +5, +6, +7 — Acidic — except amphoteric

Non-metallic oxides $\rightarrow$ Acidic except CO, NO, N_2O (neutral)
H_2O (amphoteric)
Metalloids $\rightarrow$ Acidic except amphoteric (As^{3+}, Sb^{3+})

(v) Nature of Hydroxides (A—O—H) :

$$A-O-H \begin{cases} \longrightarrow X_A - X_O \leq X_O - X_H \longrightarrow AO^- + H^+ \text{ (Acidic)} \\ \longrightarrow X_A - X_O \geq X_O - X_H \longrightarrow A^+ + OH^+ \text{ (Basic)} \end{cases}$$

EN of A decreases, X_A–X_O increases basic nature of A–O–H increases

e.g. : In NaOH

$(X_O - X_{Na} = 2.6) > (X_O - X_H = 1.4)$

Therefore, NaOH is basic in nature

e.g. : In ClOH

$(X_C - X_{Cl} = 0.5) < (X_O - X_H = 1.4)$

Therefore, ClOH is acidic in nature

e.g. : Acidic nature of HOF > HOCl > HOBr > HOI

down the group EN of halogen decreases, $X_O - X_A$ increases, Acidic Nature decreases

Exception: Acidic nature of $H_3PO_4 < H_3PO_3 < H_3PO_2$

(Based on resonance)

Q75 **Which are of the following represent the electronic configuration of the host electropositive.**

(A) $[He]2S^1$ **(B)** $[Xe]6S^2$ **(C)** $[He]2S^2$ **(D)** $[Xe]2S^2$

Sol: **B**

In a periodic table from left to right electropositive increases.

Q76 **An atom with high electronegative has:**

(A) Large size **(B) High ionisation potential**

(C) Low electro affinity **(D) Low ionisation potential**

Sol: **B**

In the periodic table.

When we go from left to right the electronegative increases and ionisation potential also increases

Q77 **Keeping in view the periodic law and the periodic table suggest which of the following elements should have the maximum electronegative character?**

(A) P **(B) As** **(C) Bi** **(D) Sb**

Sol: **A**

In the periodic table.

When we go top to bottom the electronegative character ↓(Decreases)

P
As
Sb
Bi

EN ↓

The Maximum electronegative character by phosphorus

Q78 **The outermost electronic configuration of the most electronegative element.**
(A) ns^2np^3 **(B) $ns^2\ np^4$** **(C) ns^2np^5** **(D) ns^2np^6**

Sol: **C**
In the periodic table left to right electroactivity increases (F) fluorine halogen is the outermost electronic configuration of the most electronegative element is ns^2np^5

Q79 **Two elements whose electronegative are 1.2 and 3.0 the bond formed between them would be.**
(A) Ionic **(B) Covalent** **(C) Coordinate** **(D) Metallic**

Sol: **B**
Delta electronegative less than 2.1 $\Delta \in N < 2.1$

Q80 **Electronegative is a measure of the capacity of an atoms to.**
(A) Attract electrons **(B) Attract protons**
(C) Repel electrons **(D) Repel protons**

Sol: **A**
Electronegative tendency to attract the electron pair.

Q81 **Which of the following is the most electropositive elements.**
(A) Aluminium **(B) Magnesium**
(C) Phosphorus **(D) Sulphur**

Sol: **B**

$$E.N \propto \frac{1}{E.P}$$

Electronegativity ↑ Increases then electropositive positivity decreases.

Q82 **Going from fluorine to chlorine, bromine and iodine the electronegativity.**

(A) Increases **(B) Decreases**

(C) First decreases then increases **(D) Changes randomly**

Sol: **B**

Electronegativity decreases down the groups.

Lattice Energy :

- Definition 1 :

  Amount of energy released when one mole crystal lattice is formed from its constituent gaseous ions.

  $$\underset{1\,\text{mole}}{A^+(g) + B^-(g) \longrightarrow AB(s)} \qquad (\Delta H_{LE} = \ominus ve)$$

- Definition 2 :

  Amount of energy required to break 1 mole of ionic lattice, into its constituent gaseous ions.

  $$\underset{1\,\text{mole}}{AB(s) \longrightarrow A^+(g) + B^-(g)} \qquad (\Delta H_{LE} = \oplus ve)$$

- Mathematically :

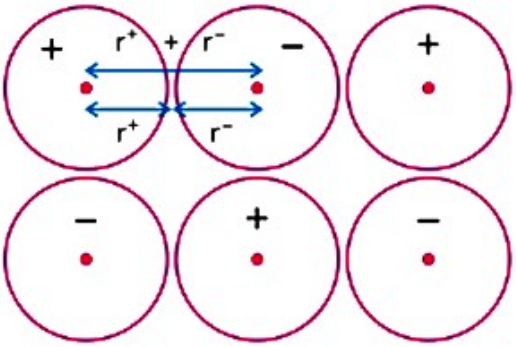
  $$E = \frac{k\,q_1\,q_2}{r^2}$$

  Where, q_1 = charge on cation

  q_2 = charge on anion

  r = internuclear distance

- Factors affecting lattice energy (L.E) :



1) Charge on ions increases, LE decreases

$LE \alpha (q_1 \times q_1)$

2) Size of ions increases, LE decreases

$$LE \alpha \frac{1}{(r^+ + r^-)}$$ (priority of charge > size)

e.g. : order of lattice energy :

1) $LiF > NaF > KF > RbF > CsF$ $(r^+ \uparrow$ L.E$\downarrow)$

2) $NaF > NaCl > NaBr > NaI$ $(r^- \uparrow$ LE$\downarrow)$

3) $NaF < MgF_2 < AlF_3$ (charge on cation$\uparrow$ LE$\downarrow$)

4) $HgF_2 < MgO < Mg_3N_2$ (charge on anion$\uparrow$ LE$\uparrow$)

5) $NaF < MgO < AlN$ (charge on ions$\uparrow$ LE$\uparrow$)

6) Mg_3N_2 Al_2O_3

$q^+ \rightarrow Mg^{2+} < Al^{3+}$

$q^- \rightarrow N^{3-} > O^{2-}$

$r^+ \rightarrow Mg^{2+} > Al^{3+}$

$r^- \rightarrow N^{3-} > O^{2-}$

$(r^+ + r^-) \rightarrow Mg_3N_2 > Al_2O_3$

$LE \rightarrow Mg_3N_2 < Al_2O_3$

Hydration Energy :

Amount of energy released when one mole gaseous ion is completely dissolved in water.

$A^+(g) \longrightarrow A^+(aq)$ $B^-(g) \longrightarrow B^-(aq)$



ΔH_{HE} of A^+ ΔH_{HE} of B^-

Factors affecting hydration energy :

1) Hydration energy $\propto$ charge of ions

2) Hydration energy $\propto \frac{1}{\text{size of ions}}$

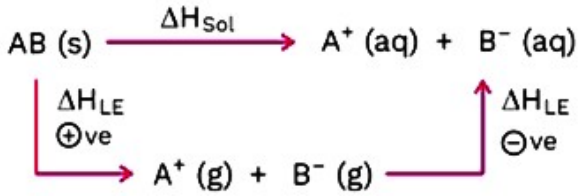
Application of HE :

1) Size of ion in aqueous medium $\propto$ HE

2) Mobility of ions in aqueous medium $\propto \frac{1}{HE}$

3) Electrical conductance $\propto \frac{1}{\text{size of ion (aq)}} \propto \frac{1}{HE}$

4) Solubility $\propto$ HE



An ionic compound is soluble if $\Delta H_{HE} > \Delta H_{LE}$

e.g., For alkali metal ions :

Size of gaseous ions : $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

Hydration energy (ΔH_{HE}) : $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

Size of hydrated ions : $Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

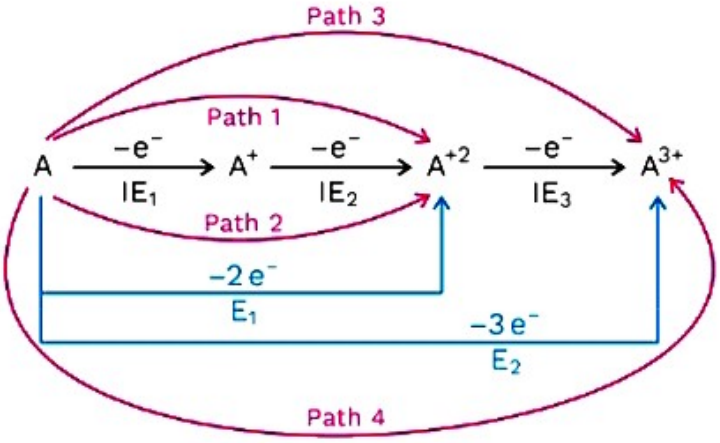
Mobility of ions (aq) : $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

Electrical conductance : $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$

Hess's Law :

Energy is a state function and not path function, that is if there are more than one path for a conversion, then energy involved in each path is same.

e.g. :



From Hess's Law : Energy of path 1 = Energy of path 2

$E_1 = IE_1 + IE_2$

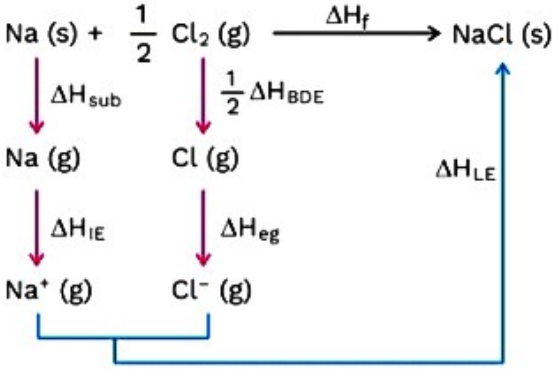
From Hess's Law : Energy of path 3 = Energy of path 4

$E_2 = IE_1 + IE_2 + IE_3$

Born Haber's Cycle :

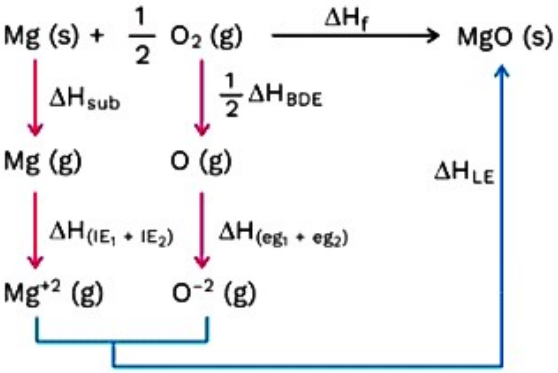
Born Haber's cycle is used to calculate Lattice energy.

For NaCl (s),



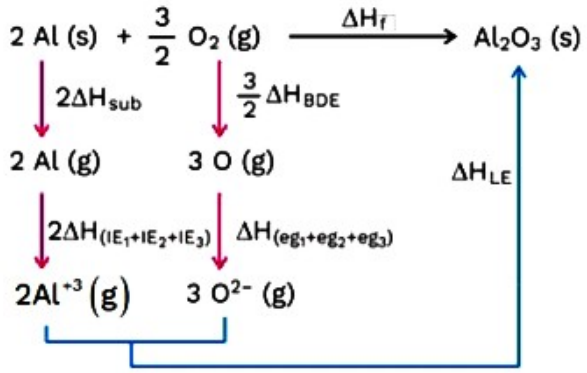
$$\Delta H_f = \Delta H_{sub} + \Delta H_{IE} + \frac{1}{2}\Delta H_{BDE} + \Delta H_{eg} + \Delta H_{LE}$$

For MgO (s),



$$\Delta H_f = \Delta H_{sub} + \Delta H_{(IE_1+IE_2)} + \frac{1}{2}\Delta H_{BDE} + \Delta H_{(eg_1+eg_2)} + \Delta H_{LE}$$

For Al_2O_3 (s),



$$\Delta H_f = 2\Delta H_{sub} + 2\Delta H_{(IE_1+IE_2+IE_3)} + \frac{3}{2}\Delta H_{BDE} + 3\Delta H_{(eg_1+eg_2+eg_3)} + \Delta H_{LE}$$

Valency:

Combining capacity of atom

Valency with respect to hydrogen

$$\underset{\text{increases}}{\xrightarrow{NaH\ MgH_2\ AlH_3\ SiH_4}} \qquad \underset{\text{decreases}}{\xrightarrow{PH_3\ H_2S\ HCl}}$$

Valency with respect to oxygen

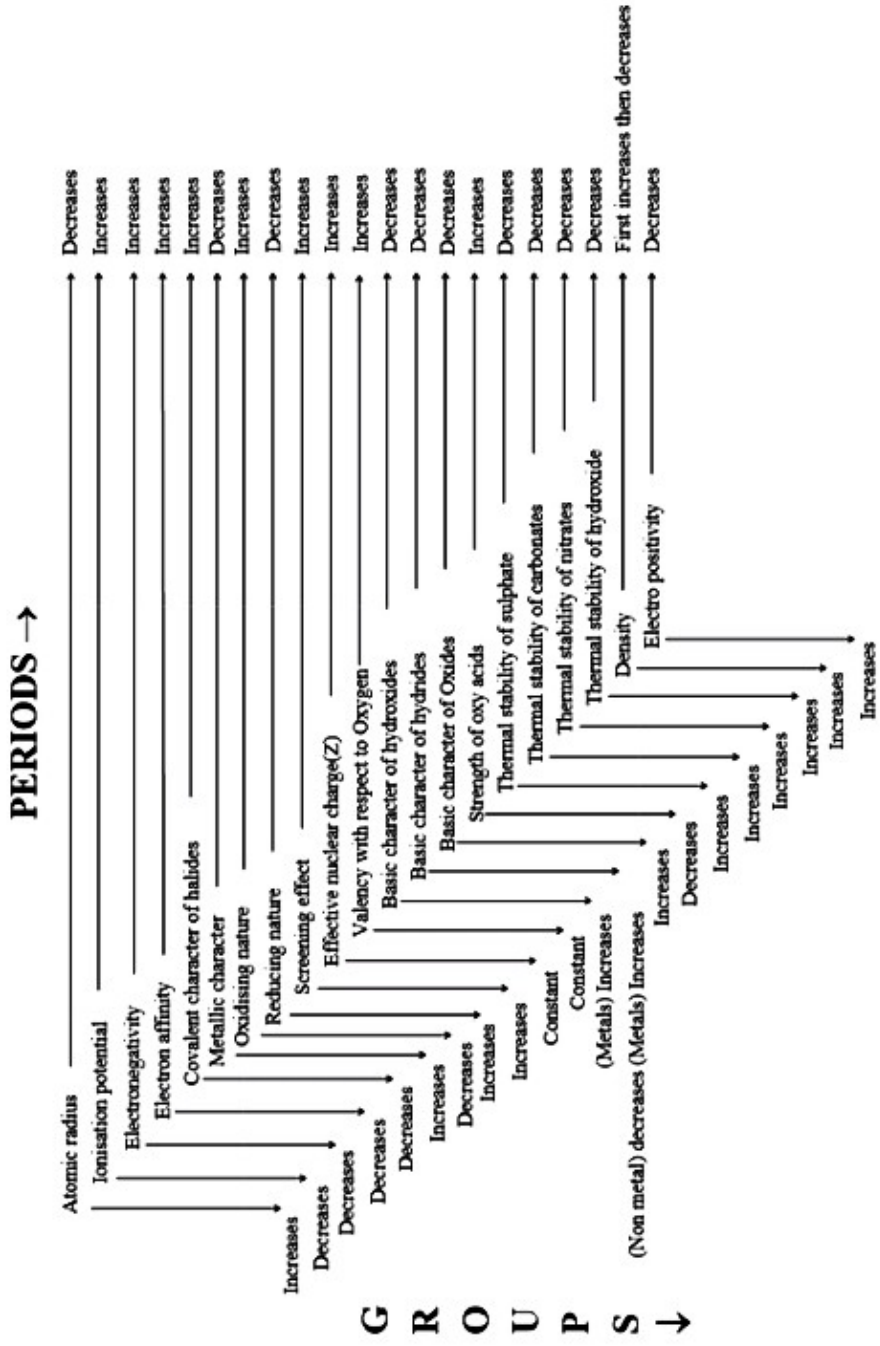
Na_2O	MgO	Al_2O_3	SiO_2	P_4O_{10}	SO_3	Cl_2O_7
1	2	3	4	5	6	7

Based on O.S.

$\overset{-7}{Cl}H_7$ not possible

$\overset{+7}{Cl_2}O_7$ possible

Some important trends-


PERIODS →
GROUPS →
Atomic radius
Ionisation potential
Electronegativity
Electron affinity
Covalent character of halides
Metallic character
Oxidising nature
Reducing nature
Screening effect
Effective nuclear charge(Z)
Valency with respect to Oxygen
Basic character of hydroxides
Basic character of hydrides
Basic character of Oxides
Strength of oxy acids
Thermal stability of sulphate
Thermal stability of carbonates
Thermal stability of nitrates
Thermal stability of hydroxide
Density
Electro positivity
Decreases
Increases
Increases
Increases
Increases
Decreases
Increases
Decreases
Increases
Decreases
Decreases
Decreases
Increases
Decreases
Decreases
Decreases
Decreases
First increases then decreases
Decreases
Increases
Decreases
Decreases
Increases
Increases
Increases
Incre

Chapter Summary



IMPORTANT FACTS TO REMEMBER

1. Lowest electronegativity	Cs
2. Highest electronegativity	F
3. Highest ionisation potential	He
4. Lowest ionisation potential	Cs
5. Lowest electron affinity	Noble gases
6. Highest electron affinity	Chlorine
7. Least electropositive element	F
8. Lowest m. pt. metal	Hg
9. Highest m. pt. and b. pt. metal	W (Tungsten)
10. Lowest m. pt. and b. pt. nonmetal	He
11. Notorious element	Hydrogen
12. Lightest element	Hydrogen
13. Smallest atomic radius	H
14. Largest atomic size	Cs
15. Largest monovalent anionic size	I^-
16. Smallest cation	H^+
17. Most electropositive element	Cs
18. Element with electronegativity after F	Oxygen next to Fluorine
19. Group containing maximum no. of gases	Zero group of gaseous elements in periodic table
20. Liquid element of radioactive nature	Fr
21. Volatile d-block elements	Zn, Cd, Hg, Uub
22. Element containing no neutron	H

Chapter Summary



IMPORTANT FACTS TO REMEMBER

23. Most abundant element on earth	Oxygen
24. Rarest element on earth	at (Astatine)
25. Most abundant metal on earth	*Al*
26. Element having maximum catenation tendency	Carbon
27. Non metal having highest m. pt., b.pt.	Carbon (Diamond)
28. Metal showing highest oxidation number	Os (+8)
29. Most electrovalent compound	CsF
30. Most stable carbonate	Cs_2CO_3
31. Strongest alkali (Hydroxide)	CsOH
32. Strongest basic oxide	Cs_2O
33. Best electricity conductor among metals	Ag
34. Best electricity conductor among non metals	graphite
35. Most poisonous element	Pu (Plutonium)
36. Liquid non metal	Br
37. Element kept in water	Phosphorus
38. Elements kept in kerosene	IA group element (except Li)
39. Element sublime on heating	I_2
40. Noble metals	Au, Pt etc.
41. Amphoteric metal	Be, Zn, Al, Sn, Pb
42. Amphoteric nonmetal	Si
43. Metalloid's elements	Si, As, Te, At, Ge, Sb, Po
44. Nonmetals having metallic lusture	Graphite, Iodine

Chapter Summary



IMPORTANT FACTS TO REMEMBER

45. Heaviest naturally occurring element	Uranium
46. Poorest conductor of electricity	Diamond
47. Hardest naturally occurring element	Diamond
48. Lightest solid metal	Li
49. 90% of Sun mass	Hydrogen
50. Amphoteric oxides	BeO, Al_2O_3, ZnO, PbO_2, SnO, SnO_2 Sb_2O_3, As_2O_3 etc.
51. Neutral oxides of nonmetals	NO, CO, H_2O, N_2O
52. Dry bleacher	O_3 (ozone)
53. Dry ice	Solid CO_2
54. Artificial explosive	TNT, RDX (Research Developed Explosive etc.)
55. Oldest known organic acid	CH_3COOH
56. First noble prize of chemistry was given to	Vant Hoff
57. Some isomorphism substances	$FeSO_4.7H_2O$(Green Vitriol), $MgSO_4.7H_2O$(Epsom Salt), $ZnSO_4.7H_2O$ (White Vitriol)
58. Some efflorescent substances	$Na_2CO_3.10H_2O$, $MgSO_4.7H_2O$ etc.
59. First man made element	Tc (Technicium)
60. Smallest period	1st (2 elements)
61. Largest period in periodic table	6th (32 elements)
62. Largest group in periodic table	3rd/III B (32 elements)
63. Most abundant d-block metal	Fe

Chapter Summary



IMPORTANT FACTS TO REMEMBER

64. Most abundant s-block metal	Ca
65. Element having maximum isotopes	Sn(10)
66. Highest density (Metals)	Ir ^{3} Os
67. Highest density (Non Metals)	Boron

Common and IUPAC names of some recently discovered elements

Atomic No.	Common names	IUPAC names
104	Rutherfordium (Rf)/Kurchatovium (Ku)	Unnilquadium (Unq)
105	Dubnium (Db)/Hahnium (Ha)	Unnilpentium (Unp)
106	Seaborgium (Sg)	Unnilhexium (Unh)
107	Bohrium (Bh)	Unnilseptium (Uns)
108	Hassium (Hs)	Unniloctium (Uno)
109	Meitnerium (Mt)	Unnilennium (Une)

Some Subjective Questions -
Organize the following in increasing order of the property indicated :

Q83 **Na^+, Mg^{2+}, O^{2-}, N^{3-} (ionic radii)**

Sol: radius $\propto \frac{1}{Z_{eff}}$

$Mg^{2+} < Na^+ < O^{2-} < N^{3-}$ (N^{3-} has greatest Z_{eff}).

Q84 **Li^+, Na^+, K^+, Rb^+ and Cs^+ (hydrated radii)**

Sol: Hydrated radii
Small cations due to higher e^- density show greater solvation with water molecules. So as that size of cation increases as it solvation decreases.

$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

Q85 **SiO_2, P_2O_5, SO_3 and Cl_2O_7 (acidic nature)**

Sol: As the size of central atom increasing as it's acidic nature increasing.

$SiO_2 < P_2O_5 < SO_3 < Cl_2O_7$

Q86 **B, Al, Ga, In(electron affinity)**

Sol: B < In < Ga < Al

Q87 **I^-, F^-, Cl^- (ionic size)**

Sol: In the group move top to bottom ionic size increased

$I^- > Cl^- > F^-$

Q88 **F^-, Cl^-, Br^-, I^- (polarizability)**

Sol: As the size of anion increases, the polarization of anion of given cation (fixed cation) increases.

$F^- < Cl^- < Br^- < I^-$

Q89 **C, F, Li, O, Cs(order of IE1)**

Sol: As the size of atom increase, the IE of atom decrease.

IE_1 $\boxed{F > O > C > Li > Cs}$

Q90 **MgO, CaO, SrO, BaO (lattice energy)**

Sol: Top to bottom lattice energy decreases

MgO > CaO > SrO > BaO

Q91 **I_2, Br_2, Cl_2, F_2 (oxidizing nature)**

Sol: Top to bottom in the group oxidizing nature decreases due standard reduction potentials of halogens are positive.

$\boxed{F_2 > Cl_2 > Br_2 > I_2}$

Q92 **Most of the compounds of the transition elements are coloured. Explain why ?**

Sol: The colour of transition metal ions arises from the excitation of e^- from the d-orbitals of lower energy to the d-orbitals of higher energy. The energy required for d-d-e^- excitation is available in the visible range.

Q93 **The atomic radii of 5d transition elements is same as 4d transition elements?**

Sol: Screening effect and nuclear charge effect which oppose each other. So that both series
(5d and 4d) these effects become nearly equal and thus, there is no change in atomic radii inspite of the fact that atomic no. increases gradually.

Q94 **Alkali metals do not form dipositive ions?**

Sol: Alkali metals have ns^1 electronic configuration so loss of one e^-, they have stable electronic configuration (full filled) so that don't form dipositive ions.

Q95 **EA1 for N < EA1 for C.**

Sol: C – 1 s^2, $2s^2 2p^2$

N – 1 s^2, $2s^2$ $2p^3$

So N has half filled electronic configuration which is most stable so EA of N less than the C.

Q96 **F_2 is more powerful oxidant than Cl_2.**

Sol: Due to greater standard reduction potential (+2.87) of F_2 so more powerful oxidant than Cl_2.

Q97 **Li is more powerful reductant than Na.**

Sol: The tendency of an element to lose an e^- in solution is measured by it's standard oxidation potential value (E^0_{ox}) since alkali metals have high E^0_{ox} values, these are strong reducing agents. However it is observed that Li is the strongest reducing agent alkali metals in solution as E^0_{ox} value of Li is maximum.

Q98 **Formation of Cl^- is exothermic but from O^- formation of O^{2-} is endothermic.**

Sol: When neutral atom accepts an e^-, then process is exothermic.

$Cl \rightarrow Cl^-$

If a negative ion accepts an e^-, then process is endothermic due to repulsion of negative ion charge and e^-.

Q99 **First I.P. of B is less than Be ?**

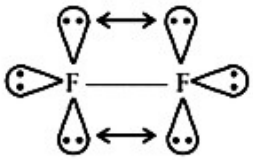
Sol: Because Be has stable configuration ($1s^2 2s^2$) whereas B has ($1s^2 2s^2 2p^1$). So B allow to remove an e^- the own outer most orbital, then first IP of B is less than Be.

Q100 **Electron affinity of 'Cl' is more than 'F' ?**

Sol: F has small size and high electron density so upcoming e^- is repelled whereas Cl has greater size therefore EA of Cl is greater than F.

Q101 **Bond length of F_2 is shorter than that of Cl_2 but F–F bond is weaker bond than Cl – Cl ?**

Sol: lp-lp repulsion force dominated more than bp-bp attraction force so F-F bond weaker and greater electronegativity of F so F-F bond length shorter than Cl-Cl bond length. Whereas in Cl-Cl bond lp-lp repulsion less due to e^- present in 3rd shell.



Repulsion

Q102 **Stability of +3 state in nitrogen family increases from top to bottom. ?**

Sol: As we move top to bottom, the atomic no. increases so d-orbitals came in shell so p-orbitals e^- moving to d-orbitals hence oxidation state increases.

Q103 **In group I A (alkali metals) only Li+ can form complex ?**

Sol: Li metal ion have small size, high charge density and empty orbitals to accommodate electron pair of ligand. Due to smallest size only lithium ion can form complex ion other IA metals not show.

Q104 **Amongst LiF and LiI which has more covalent character.**

Sol: When small size of cation and bigger size of anion have greater covalent character. So LiI – has both function so this compound more covalent than LiF.

Q105 **NaBr has M.P. greater than that of $MgBr_2$. Why?**

Sol: More ionic character in NaBr

Q106 **The atomic radius decrease along the period but inert gases has maximum atomic radius in a period.**

Sol: Due to increases Zeffective so atomic radius decreases along the period.

Atomic radii $\propto \frac{1}{Z_{eff}}$

So inert gases have greatest radius in the period.

Q107 **M.P of $CaCl_2$ is 782ºC but M.P of $HgCl_2$ is 276ºC. Why ?**

Sol: More ionic character in $CaCl_2$

Q108 **Calculate the electronegativity of fluorine from the following data :**

Sol: EH–H=104.2 kcal mol–1, EF–F= 36.6 kcal mol–1

EH–F = 134.6 kcal mol–1, XH =2.1.

$X_A - X_B = 0.208 \{E_{A-B} - [E_{A-A} \times E_{B-B}]^{1/2}]^{1/2}$

$X_F - X_H = 0.208 [E_{H-F} - [E_{F-F} \times E_{H-H})^{1/2}]^{1/2}$

$X_F - 2.1 = 0.208 [134.6 - (104.2 \times 36.6)^{1/2}]^{1/2}$

$X_F = 0.208 [134.6 - (134.6 - 62)^{1/2}]^{1/2} + 2.1$

$= 0.208 (134.6 - 62)^{1/2} + 2.1$

$= 0.208 \times 8.5 + 2.1$

$X_F = 3.87$

Q109 **Find out percentage ionic character of HF according to Pauling's equations.**

Sol: 37.8 and 44.21

Q110 **The IE values of $Al(g) \rightarrow Al^{+} + e$ is 577.5 $kJmol^{-1}$ and ΔH for $Al(g) \rightarrow Al^{+3}(g) + 3e$ is 5140 $kJ\ mol^{-1}$. If second and third IE values are in the ratio 2 : 3. Calculate IE_2 and IE_3.**

Sol: IE_2 = 18.25 kJ/mole, IE_3 = 2737.5 kJ/mole

Q111 **How many chlorine atoms will be ionised $Cl \rightarrow Cl^{+} + e^{-}$ by the energy released from the process $Cl + e^{-} \rightarrow Cl^{-}$ for 6.02×10^{23} atoms (I.E. for Cl = 1250 kJ $mole^{-1}$ and E.A. = 350 kJ $mole^{-1}$)**

Sol:

$$Al_{(g)} \xrightarrow{\Delta E=577.5} Al^{+} + e^{-} \longrightarrow Al^{+2} \longrightarrow Al^{+3}$$

$$\Delta E_2 = 5140 \text{ kJ/mole}$$

$\Delta E_3 = \Delta E_2 - \Delta E_1$

$= 5140 - 577.5 = 4562.50$

IE_2 and IE_3 have ratio 2 : 5

$$IE_2 = \frac{2}{2+3} \times 4562.50 = 1825 \text{ kJ/mole}$$

$$IE_3 = \frac{3}{2+3} \times 4562.50 = 2737.5 \text{ kJ/mole.}$$

Q112 **For the gaseous reaction,**
$K_{(g)} + F_{(g)} \rightarrow K^{+}_{(g)} + F^{-}_{(g)}$, ΔH was calculated to be 19 kcal/mol under conditions where the cations and anions were prevented by electrostatic separation from combining with each other. The ionisation energy of K is 4.3 eV. What is the electron affinity of F?(in eV)

Sol:

$K \rightarrow K^{+} + e^{-}$ $\quad \Delta E_1 = 4.3$ eV

$F + e^{-} \rightarrow F^{-}$ $\quad \Delta E_2 = -E$ eV

$$\frac{19.0}{23.06} = \Delta E_1 - \Delta E_3 = 4.3 - E$$

$0.82 = 4.3 - E$

$E = 3.48$

Q113 **Calculate Z_{eff} for**
(i) 3d electron of Cu.
(ii) 4s electron of Cu.

Sol: (i) $Ze_{ff} = Z - \sigma = 29 - (9 \times 0.35 + 18 \times 1 = 7.85)$
(ii) $Z_{eff} = Z - \sigma = 29 - (18 \times 0.85 + 10 \times 1) = 3.7$

Q114 **Find H–F bond length if covalent radius of H is 0.37 Å and that of fluorine is 0.71 Å.**

Sol: Bond length = $r_c^+ + r_c^- = 0.37 + 0.71 = 1.08$ Å

Q115 **Calculate E.N. of Si. (Covalent radius of Si = 1.175 Å)**

Sol: (E.N.)Allred Rochow = $0.744 + \frac{0.359}{r^2} Z_{eff}$

Z_{eff} of $Si^+ \Rightarrow Si = 1s^2 2s^2 2p^6 3s^2 3p^2$

$\therefore \quad Z_{eff} = 14 - (3 \times 0.35 + 8 \times 0.85 + 2 \times 1)$
$= 4.15$

$\therefore \quad E.N = 0.744 + \frac{0.359}{(1.175)} \times 4.15$

$= 1.823$

Q116 **If I.E. of F^- is 328 kJ/mol and E.A. of F^+ is 1681 kJ/mol, then find out E.N. of F at Pauling's scale ?**

Sol: I.E. of F^- = 328 kJ/mol $\Rightarrow$ E.A. of F = 328 kJ/mol E.A. = 3.4 ev
E.A of F^- = 1681 kJ/mol $\Rightarrow$ I.E of F = 1681 kJ/mol = 17.45 sec

$\therefore \quad (E.N)_m = \frac{IF_2 + F_2A}{2} = 10.43$

$\therefore \quad (E.N.)_p = \frac{(E.N.)_m}{2.8} = 3.72$